Supporting Information for "Physically Meaningful Modeling of Dst Evolution From Symbolic Regression"

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1 Unconstrained primitive equations

1.1 Equations

$$\frac{dDst}{dt} = \frac{0.0193310740849303(BN)^{1/4}}{N + 1.9690949} \times \Big[-V(B_s + 1.4056387)(0.001023190Dst + 0.04994203N + 0.098340596568647) + (N + 1.9690949)(-Dst + N + 26.93975)\Big] \tag{6}$$

$$\frac{dDst}{dt} = -B_s(0.0024323284B_sN + 0.00550733965633361Dst + 0.0024323284V) - 0.052771024Dst \tag{7}$$

$$\frac{dDst}{dt} = -B_s(0.0025749283B_sN + 0.00633902208966901Dst + 0.0025749283V) - 0.04895255Dst + 0.04895255N \tag{8}$$

1.2 Metrics per equation per storm

Table 1: RMSE, MAE, R^2, and BFE for the unconstrained primitive-feature equations (Eqs. 6–8) across the 20 test storms.

Storm	Equation 6				Equation 7				Equation 8			
	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE
54	9.176	6.120	0.792	11.066	11.585	8.862	0.669	16.184	12.200	9.334	0.633	15.867
55	14.873	11.446	0.817	15.958	16.240	13.384	0.782	16.825	16.102	13.040	0.786	16.417
56	16.063	10.936	0.434	19.312	10.639	8.542	0.752	12.855	11.069	8.717	0.731	13.648
57	9.186	7.115	0.785	8.332	9.657	7.793	0.763	8.812	10.061	8.250	0.743	8.561
58	8.458	6.020	0.827	13.608	9.013	6.801	0.804	13.498	8.467	6.228	0.827	13.939
59	18.585	15.739	0.652	11.690	13.676	11.807	0.812	11.507	14.563	12.563	0.786	11.121
60	23.885	17.550	0.591	35.776	17.859	14.196	0.772	20.260	18.325	14.755	0.760	19.980
61	15.669	9.179	0.854	25.508	11.922	9.179	0.915	16.535	11.827	8.774	0.917	16.329
62	12.475	10.340	0.638	10.320	13.974	11.807	0.546	12.900	13.457	11.221	0.579	12.298
63	10.747	7.477	0.916	16.027	17.042	13.963	0.789	21.014	16.559	13.502	0.801	21.085
64	8.406	6.356	0.864	9.423	9.881	7.914	0.812	11.855	9.710	7.799	0.818	11.414
65	15.354	12.149	0.575	19.358	16.517	11.846	0.508	13.718	16.140	11.536	0.530	13.079
66	15.705	11.809	0.662	21.214	20.919	16.088	0.400	26.436	21.019	15.917	0.394	26.715
67	8.765	6.759	0.863	11.204	9.507	7.260	0.839	11.657	9.433	7.443	0.842	11.263
68	21.684	15.933	0.967	21.878	23.371	18.995	0.962	23.228	21.083	16.912	0.969	20.970
69	11.824	8.939	0.754	12.993	12.434	9.530	0.728	13.330	11.135	8.821	0.782	12.451
70	18.157	13.536	0.810	19.354	21.146	15.651	0.743	23.785	19.434	14.307	0.783	23.072
71	12.620	9.903	0.881	14.512	13.001	9.582	0.874	15.370	13.144	10.039	0.871	15.487
72	30.306	22.817	0.859	37.042	26.312	16.568	0.894	31.583	26.719	17.017	0.890	32.485
73	10.719	8.162	0.802	10.375	11.874	9.665	0.757	10.326	10.251	7.983	0.819	8.856
Mean	14.633	10.914	0.767	17.247	14.828	11.472	0.756	16.584	14.535	11.208	0.763	16.252
Global	15.078	10.593	0.868	30.522	15.268	11.297	0.864	26.809	14.907	11.032	0.871	26.059

1.3 Figures per storm

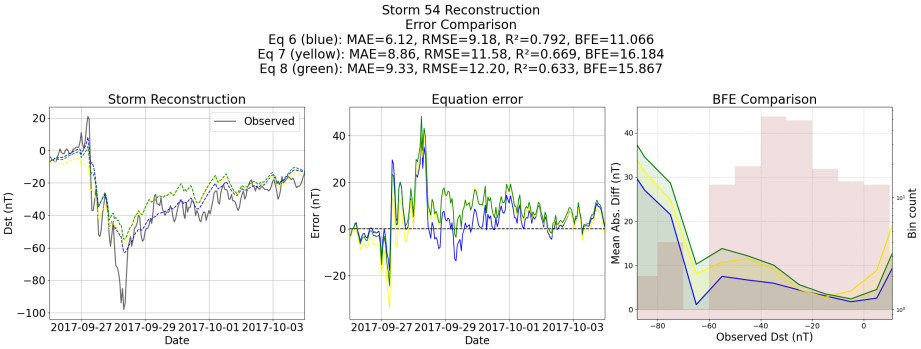

Figure 1: Reconstruction of storm 54 using the Equations generated from the unconstrained symbolic regression with the primitive features

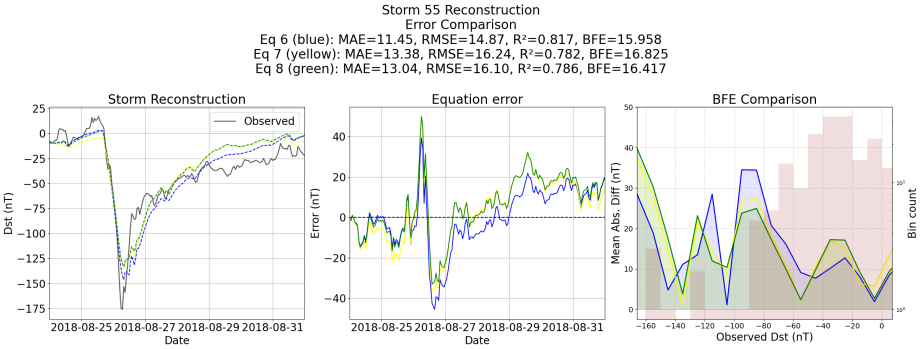

Figure 2: Reconstruction of storm 55 using the Equations generated from the unconstrained symbolic regression with the primitive features

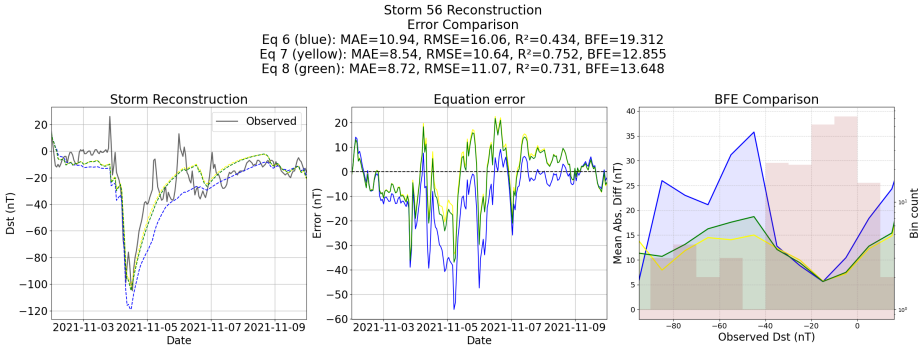

Figure 3: Reconstruction of storm 56 using the Equations generated from the unconstrained symbolic regression with the primitive features

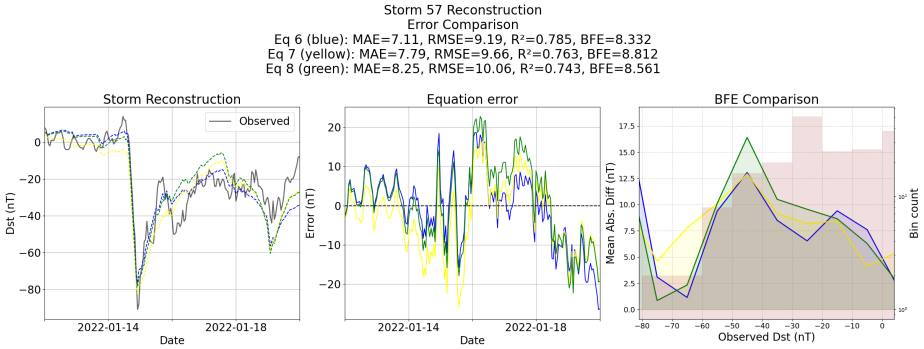

Figure 4: Reconstruction of storm 57 using the Equations generated from the unconstrained symbolic regression with the primitive features

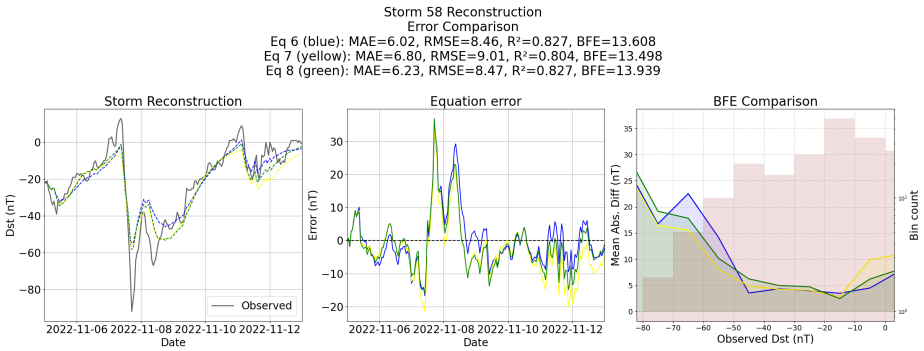

Figure 5: Reconstruction of storm 58 using the Equations generated from the unconstrained symbolic regression with the primitive features

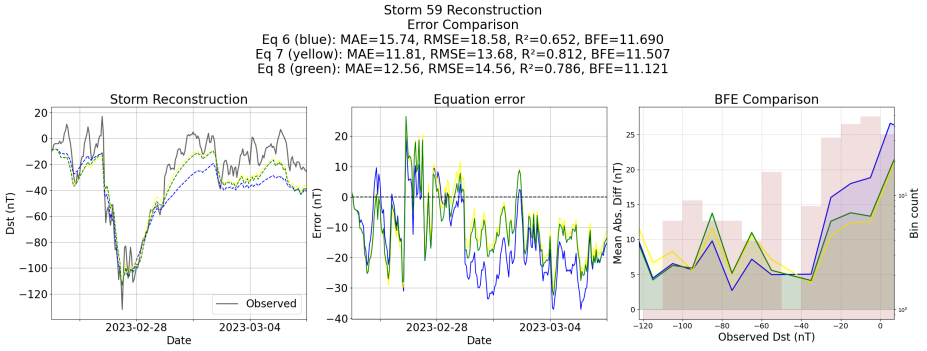

Figure 6: Reconstruction of storm 59 using the Equations generated from the unconstrained symbolic regression with the primitive features

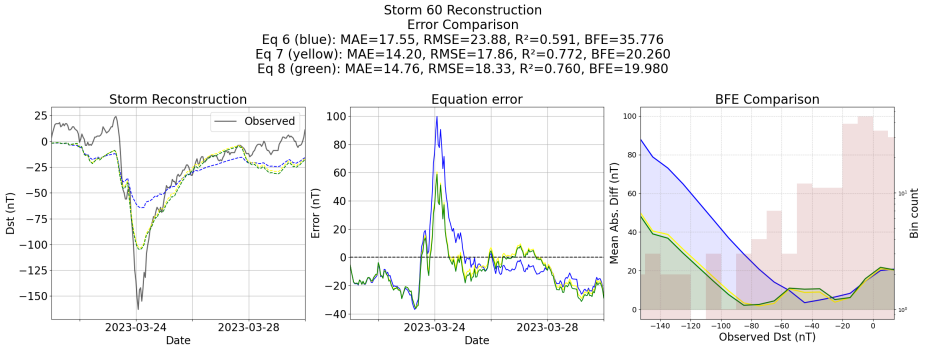

Figure 7: Reconstruction of storm 60 using the Equations generated from the unconstrained symbolic regression with the primitive features

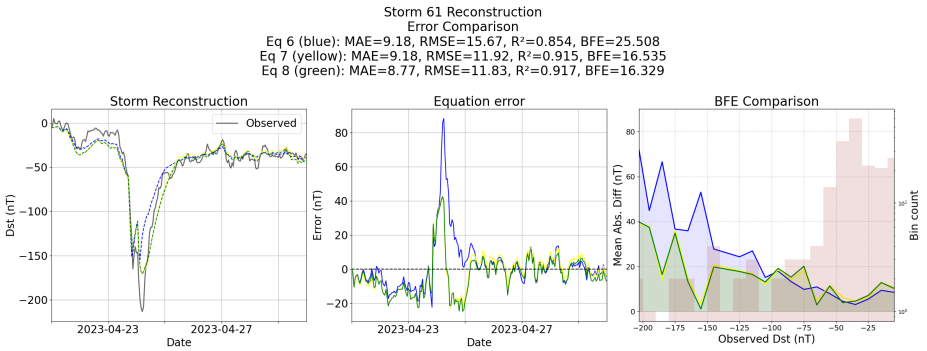

Figure 8: Reconstruction of storm 61 using the Equations generated from the unconstrained symbolic regression with the primitive features

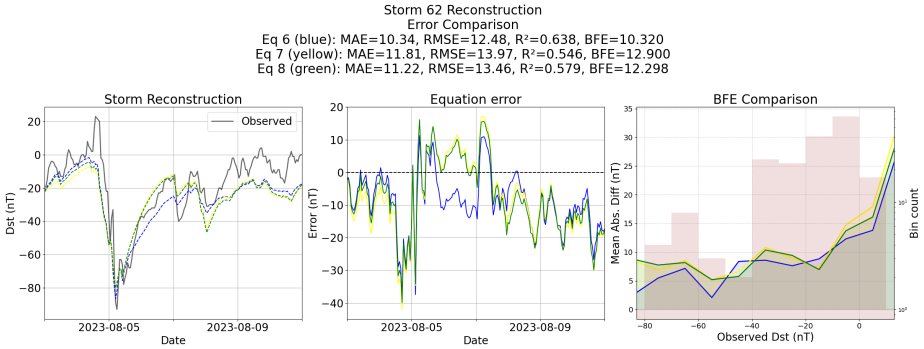

Figure 9: Reconstruction of storm 62 using the Equations generated from the unconstrained symbolic regression with the primitive features

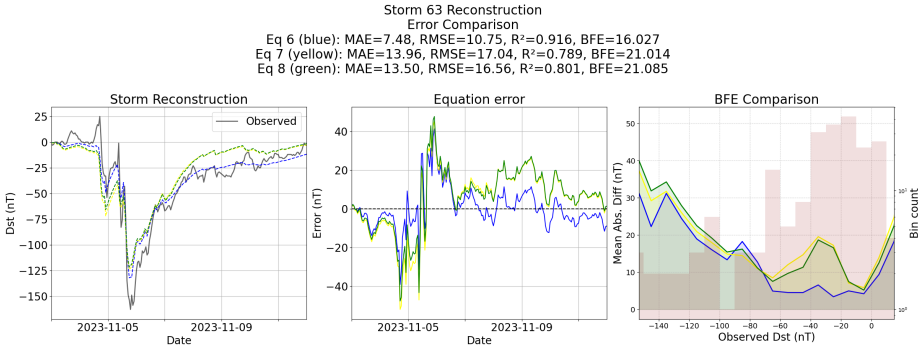

Figure 10: Reconstruction of storm 63 using the Equations generated from the unconstrained symbolic regression with the primitive features

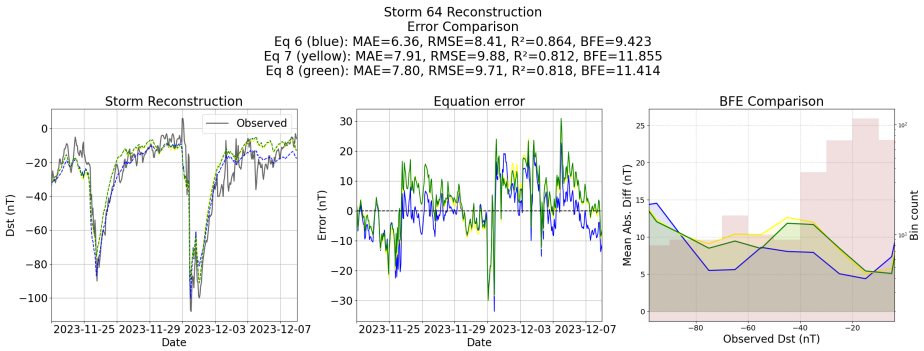

Figure 11: Reconstruction of storm 64 using the Equations generated from the unconstrained symbolic regression with the primitive features

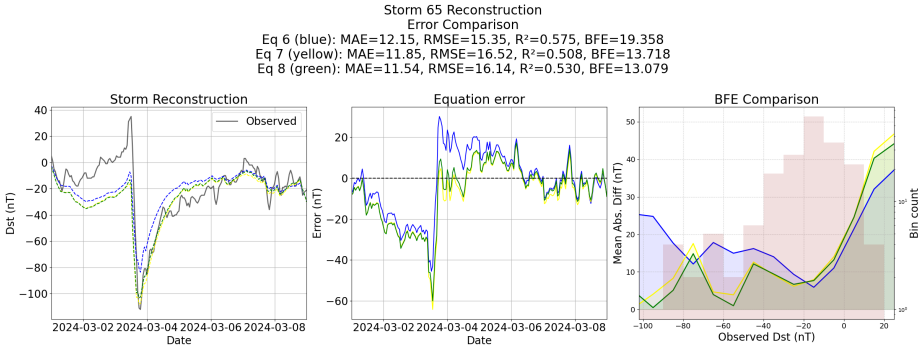

Figure 12: Reconstruction of storm 65 using the Equations generated from the unconstrained symbolic regression with the primitive features

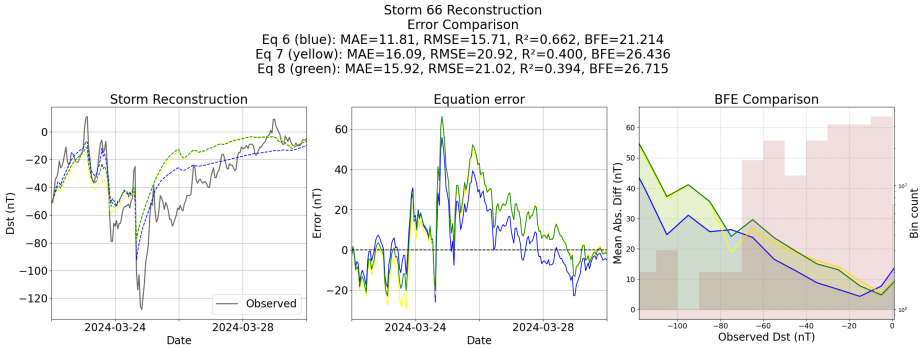

Figure 13: Reconstruction of storm 66 using the Equations generated from the unconstrained symbolic regression with the primitive features

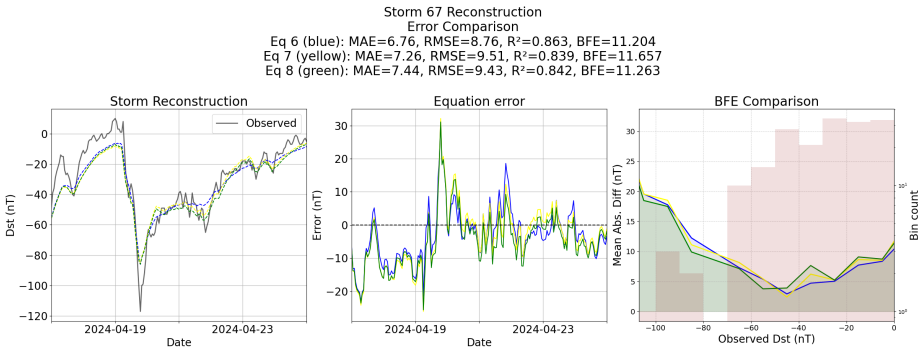

Figure 14: Reconstruction of storm 67 using the Equations generated from the unconstrained symbolic regression with the primitive features

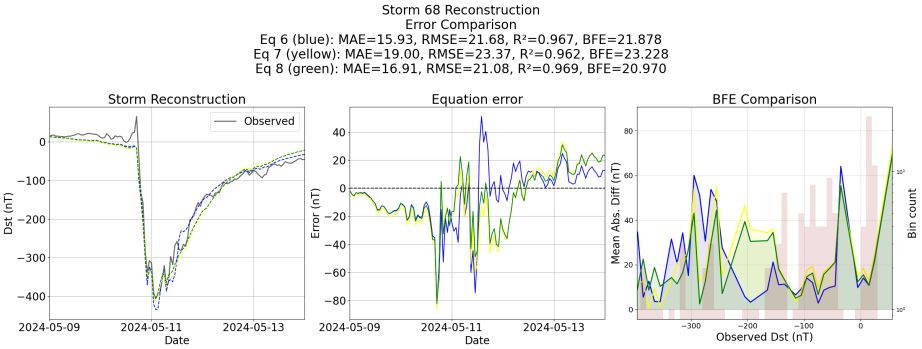

Figure 15: Reconstruction of storm 68 using the Equations generated from the unconstrained symbolic regression with the primitive features

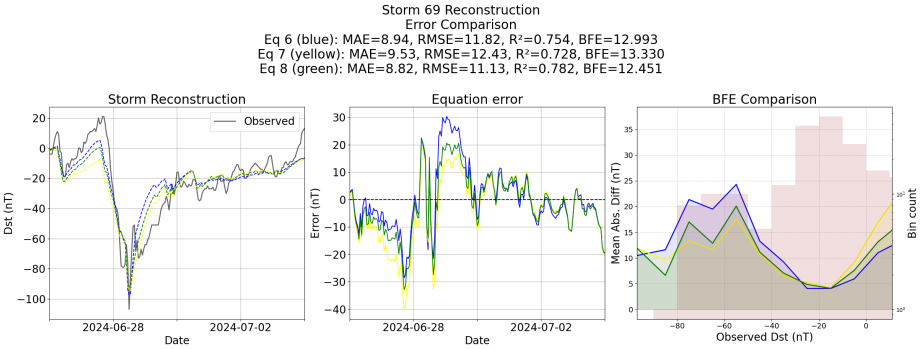

Figure 16: Reconstruction of storm 69 using the Equations generated from the unconstrained symbolic regression with the primitive features

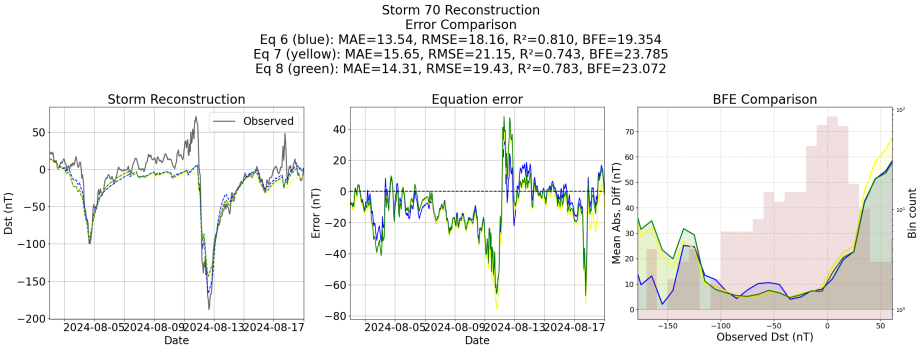

Figure 17: Reconstruction of storm 70 using the Equations generated from the unconstrained symbolic regression with the primitive features

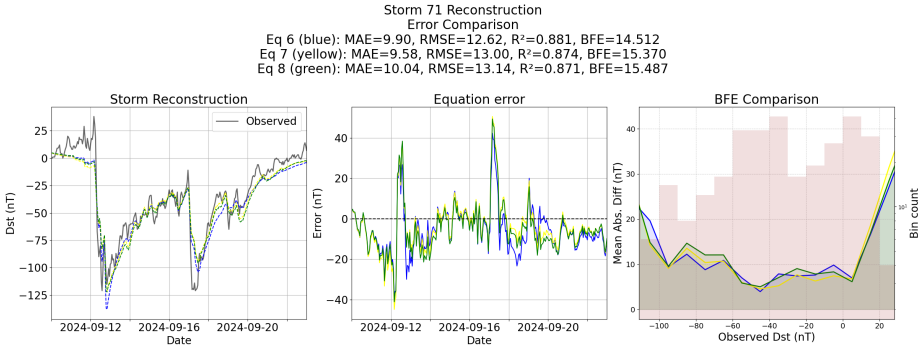

Figure 18: Reconstruction of storm 71 using the Equations generated from the unconstrained symbolic regression with the primitive features

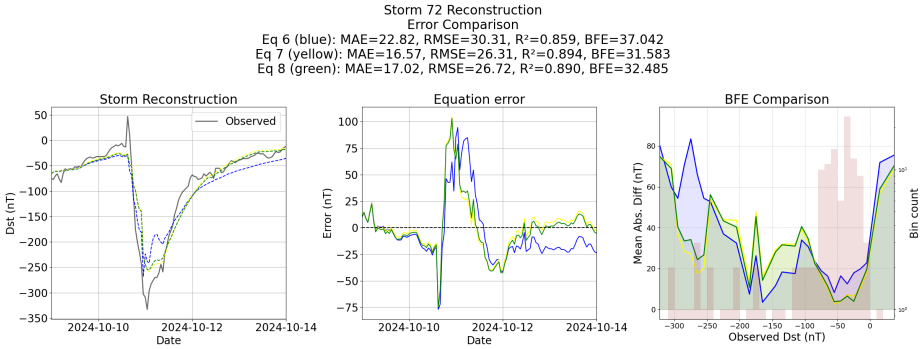

Figure 19: Reconstruction of storm 72 using the Equations generated from the unconstrained symbolic regression with the primitive features

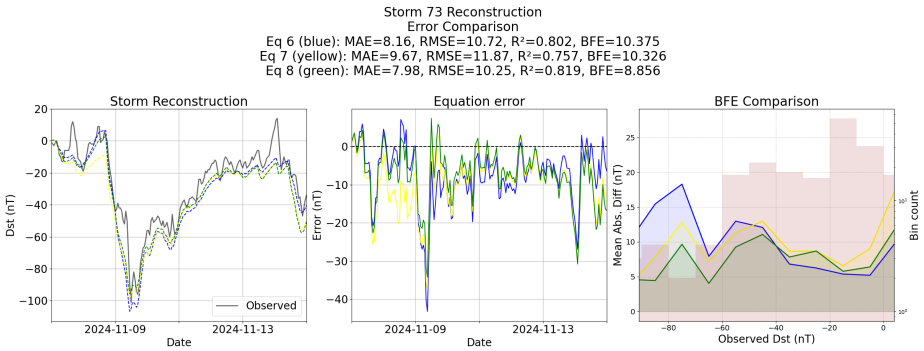

Figure 20: Reconstruction of storm 73 using the Equations generated from the unconstrained symbolic regression with the primitive features

2 Unconstrained derived equations

2.1 Equations

$$\frac{dDst}{dt} = \frac{VB_s\sqrt{P_{dyn} + 0.887826} - (0.6033792Dst - 13.5283602161664)^2}{(Dst - 825.26373)} \tag{9}$$

$$\frac{dDst}{dt} = -0.0009785303VB_s\sqrt{P_{dyn} + (0.008792949Dst + 1.9299145)^2} + (0.01791331Dst - 0.60905385)^2 \tag{10}$$

$$\frac{dDst}{dt} = -0.001091036VB_s\sqrt{P_{dyn} + 1.1339003} + (0.018421166Dst - 0.5820395)^2 \tag{11}$$

$$\frac{dDst}{dt} = -0.0011102495VB_s\sqrt{P_{dyn} + 1.0975075} + 0.00036096844(Dst - 29.48951)^2 \tag{12}$$

2.2 Metrics per equation per storm

Table 2: RMSE, MAE, R^2, and BFE for the unconstrained derived-feature equations (Eqs. 9–12) across the 20 test storms.

Storm	Equation 9				Equation 10				Equation 11				Equation 12			
	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE
54	8.065	6.121	0.839	11.309	8.315	6.189	0.829	11.418	9.009	7.113	0.800	11.572	8.779	6.908	0.810	11.485
55	15.909	12.481	0.791	18.406	16.490	12.971	0.775	18.669	17.375	13.732	0.750	19.559	17.008	13.515	0.761	19.160
56	12.655	9.273	0.648	15.835	12.133	9.311	0.677	14.531	13.364	11.015	0.608	15.722	13.103	10.590	0.623	15.542
57	10.006	7.705	0.745	10.435	10.130	8.055	0.739	11.340	9.541	7.610	0.769	12.026	9.484	7.530	0.771	11.574
58	9.828	7.555	0.767	14.806	9.422	7.335	0.786	14.756	9.192	6.388	0.796	15.626	9.199	6.491	0.796	15.365
59	13.735	11.746	0.810	9.731	13.852	11.906	0.807	10.250	12.169	10.188	0.851	9.420	12.340	10.402	0.847	9.421
60	19.322	13.056	0.733	29.310	17.574	12.932	0.779	24.013	18.267	11.955	0.761	28.251	18.338	12.103	0.759	28.237
61	10.939	7.963	0.929	14.228	12.773	8.617	0.903	17.512	10.705	8.288	0.932	14.329	10.646	8.227	0.932	13.886
62	12.966	10.539	0.609	12.925	13.310	11.045	0.588	12.627	11.351	9.077	0.700	12.941	11.643	9.346	0.685	12.911
63	13.845	10.907	0.861	17.905	15.512	12.494	0.825	20.097	14.786	12.001	0.841	17.960	14.659	11.928	0.844	18.078
64	9.289	7.052	0.834	10.507	9.932	7.741	0.810	11.165	10.060	7.945	0.805	11.160	9.894	7.756	0.812	11.043
65	14.800	10.437	0.605	15.901	16.276	11.645	0.522	17.600	14.033	10.198	0.645	16.576	14.165	10.196	0.638	16.427
66	16.904	13.666	0.608	18.624	17.967	14.870	0.557	20.191	18.102	15.283	0.550	19.895	17.959	15.043	0.557	19.767
67	11.494	8.584	0.765	15.539	10.546	7.836	0.802	13.115	11.316	8.546	0.772	16.146	11.253	8.495	0.775	15.909
68	24.451	18.648	0.959	24.482	25.056	19.165	0.956	25.588	28.616	20.918	0.943	30.788	28.429	21.074	0.944	29.892
69	11.506	8.354	0.767	12.459	11.111	8.204	0.783	12.166	11.001	8.271	0.787	12.135	11.079	8.242	0.784	12.197
70	19.571	14.622	0.780	20.840	18.575	13.664	0.802	21.490	16.946	12.186	0.835	19.225	17.372	12.555	0.826	19.343
71	11.556	8.322	0.900	13.533	11.495	8.147	0.901	13.167	11.653	7.737	0.899	13.215	11.517	7.755	0.901	13.191
72	16.534	11.844	0.958	18.784	17.477	11.769	0.953	17.463	18.094	13.083	0.950	22.067	17.264	12.543	0.954	19.846
73	13.047	11.055	0.707	11.829	12.781	11.077	0.719	11.474	11.556	9.456	0.770	10.800	11.685	9.578	0.765	10.838
Mean	13.821	10.497	0.781	15.869	14.036	10.749	0.776	15.932	13.857	10.549	0.788	16.471	13.791	10.514	0.789	16.206
Global	14.186	10.359	0.883	25.031	14.303	10.564	0.881	26.400	14.069	10.238	0.885	29.234	14.038	10.225	0.885	28.140

2.3 Figures per storm

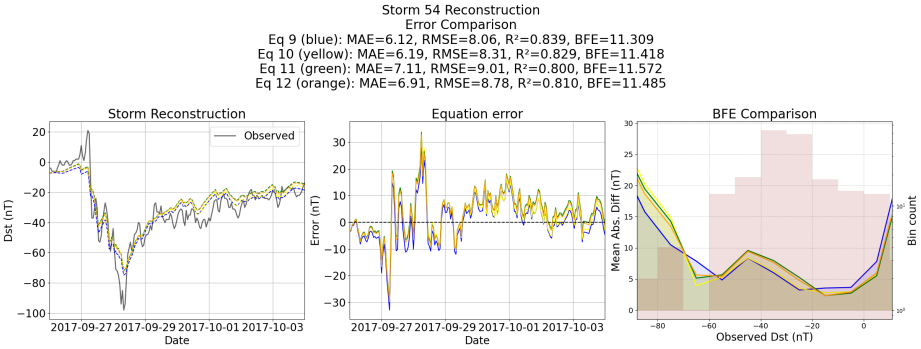

Figure 21: Reconstruction of storm 54 using the Equations generated from the unconstrained symbolic regression with the derived features

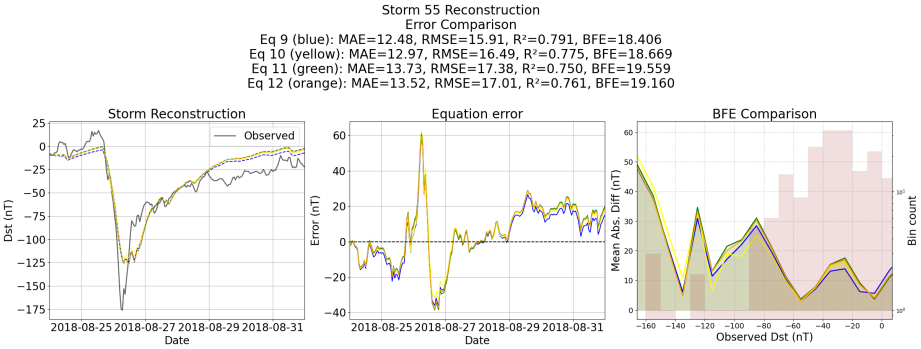

Figure 22: Reconstruction of storm 55 using the Equations generated from the unconstrained symbolic regression with the derived features

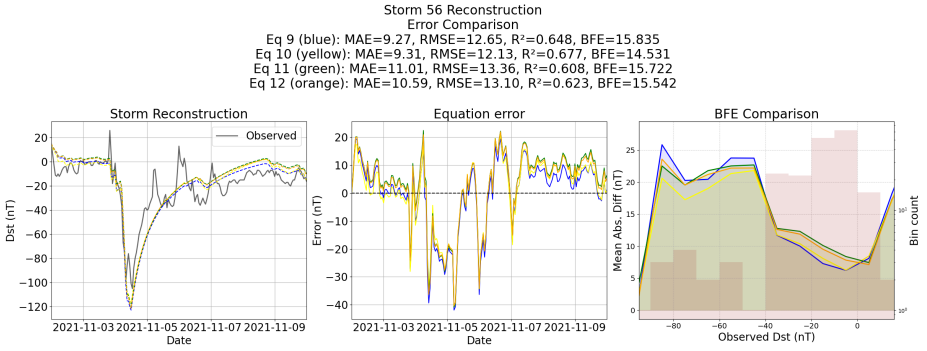

Figure 23: Reconstruction of storm 56 using the Equations generated from the unconstrained symbolic regression with the derived features

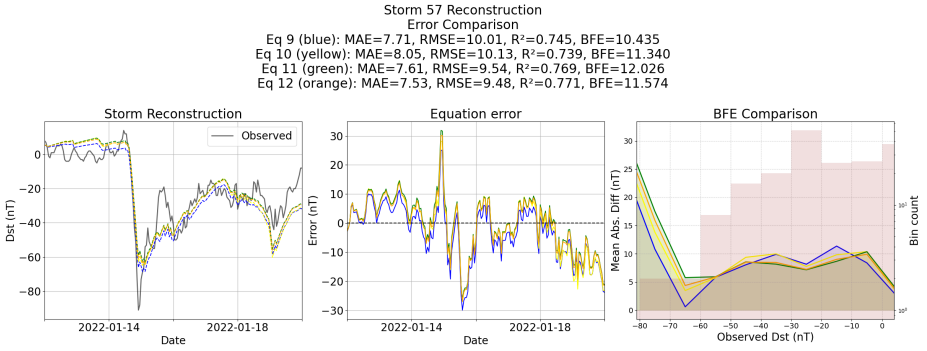

Figure 24: Reconstruction of storm 57 using the Equations generated from the unconstrained symbolic regression with the derived features

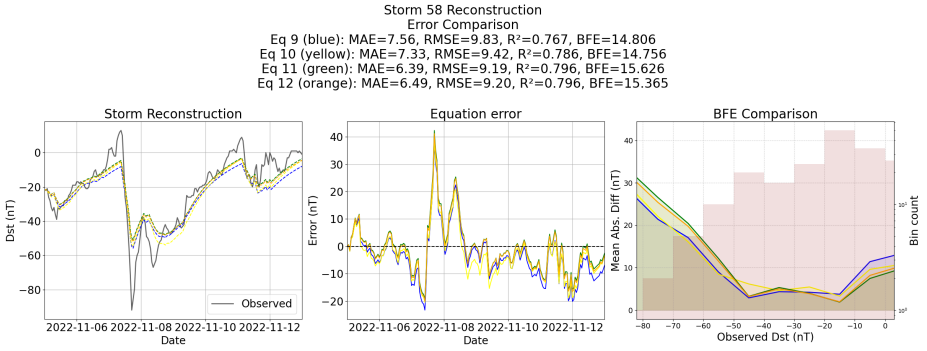

Figure 25: Reconstruction of storm 58 using the Equations generated from the unconstrained symbolic regression with the derived features

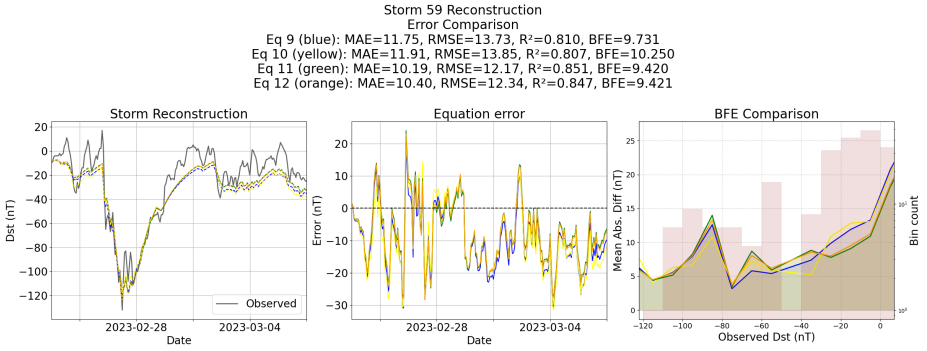

Figure 26: Reconstruction of storm 59 using the Equations generated from the unconstrained symbolic regression with the derived features

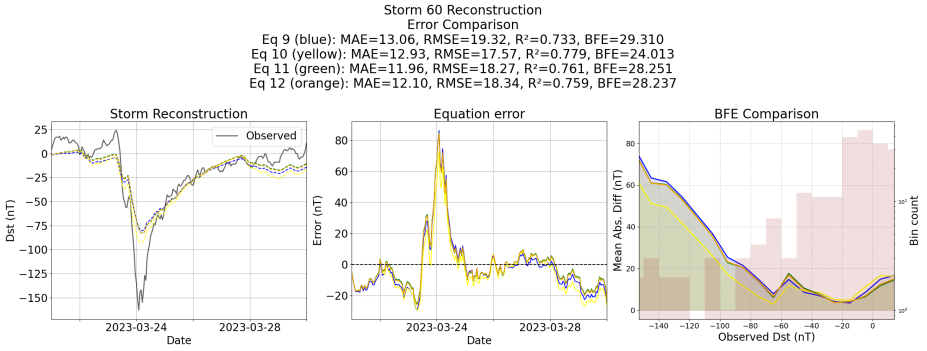

Figure 27: Reconstruction of storm 60 using the Equations generated from the unconstrained symbolic regression with the derived features

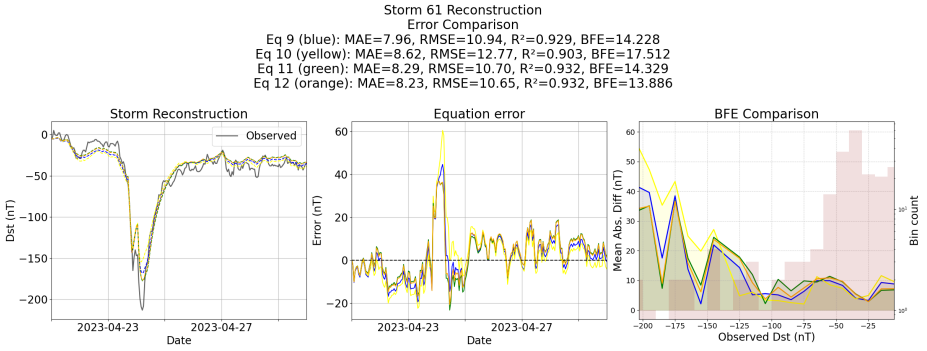

Figure 28: Reconstruction of storm 61 using the Equations generated from the unconstrained symbolic regression with the derived features

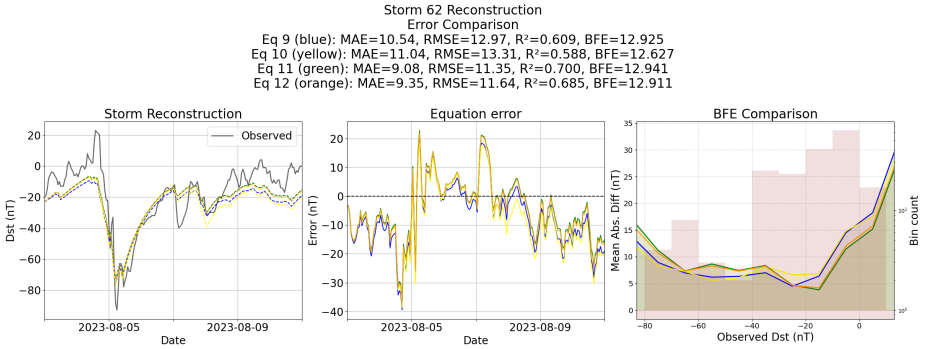

Figure 29: Reconstruction of storm 62 using the Equations generated from the unconstrained symbolic regression with the derived features

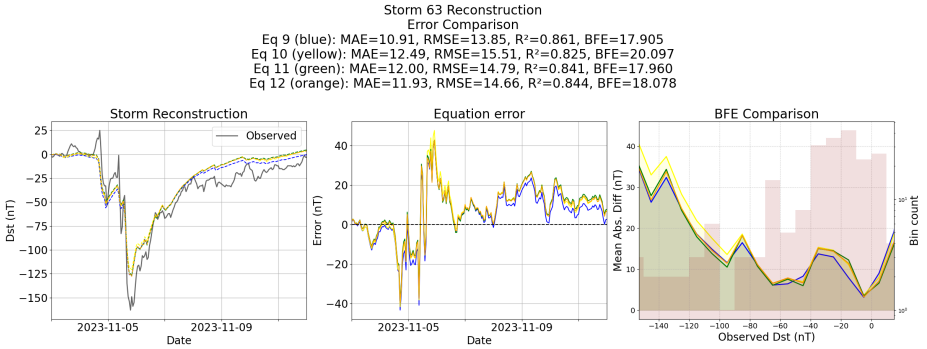

Figure 30: Reconstruction of storm 63 using the Equations generated from the unconstrained symbolic regression with the derived features

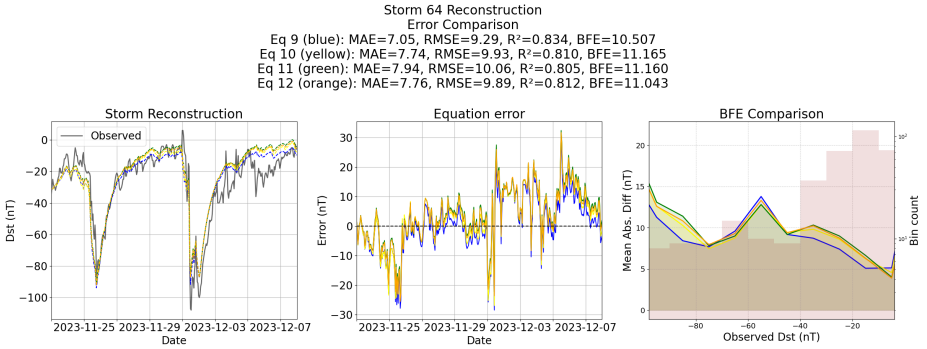

Figure 31: Reconstruction of storm 64 using the Equations generated from the unconstrained symbolic regression with the derived features

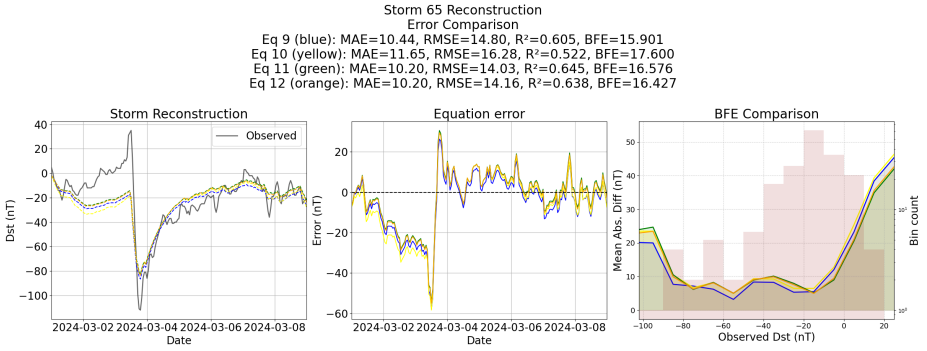

Figure 32: Reconstruction of storm 65 using the Equations generated from the unconstrained symbolic regression with the derived features

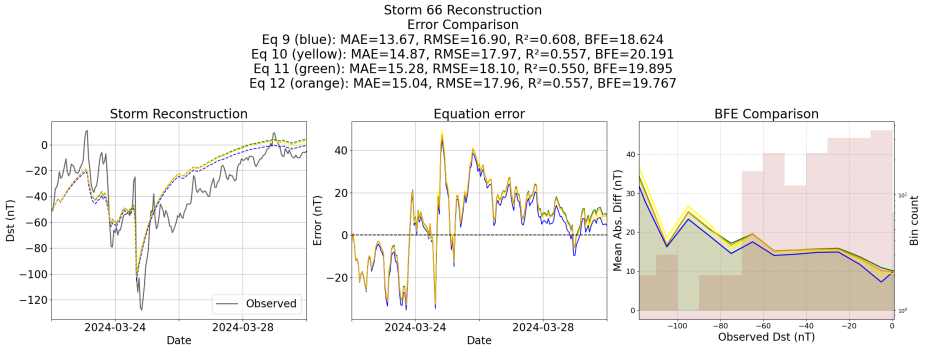

Figure 33: Reconstruction of storm 66 using the Equations generated from the unconstrained symbolic regression with the derived features

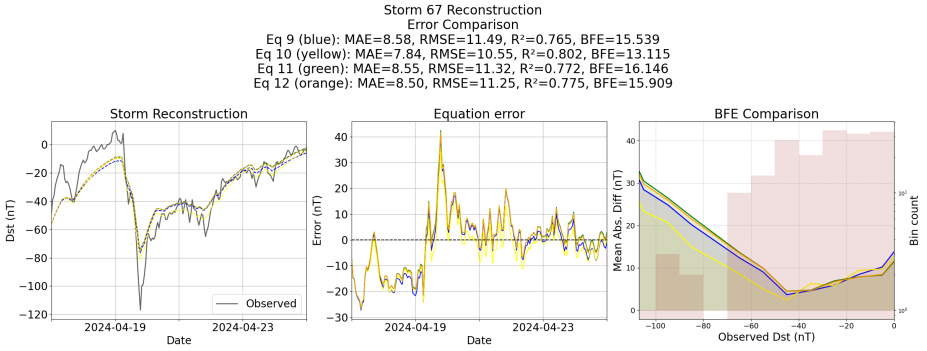

Figure 34: Reconstruction of storm 67 using the Equations generated from the unconstrained symbolic regression with the derived features

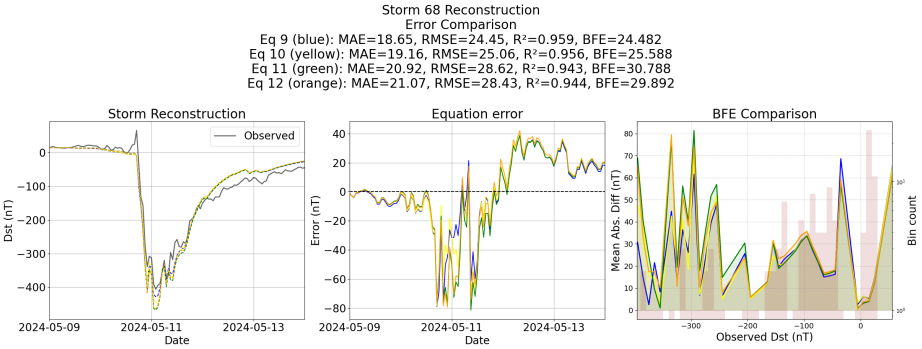

Figure 35: Reconstruction of storm 68 using the Equations generated from the unconstrained symbolic regression with the derived features

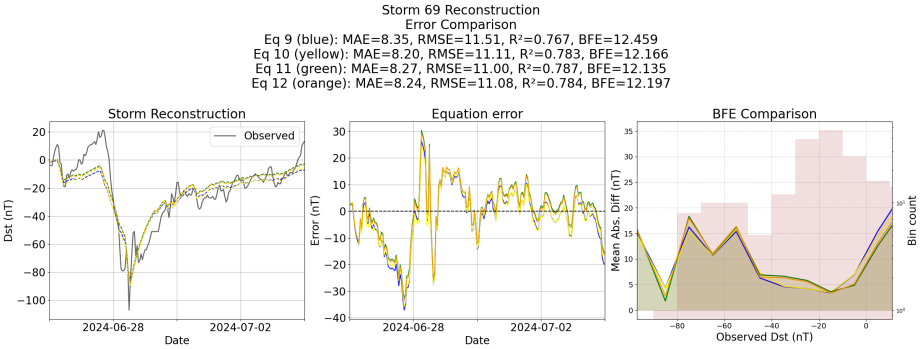

Figure 36: Reconstruction of storm 69 using the Equations generated from the unconstrained symbolic regression with the derived features

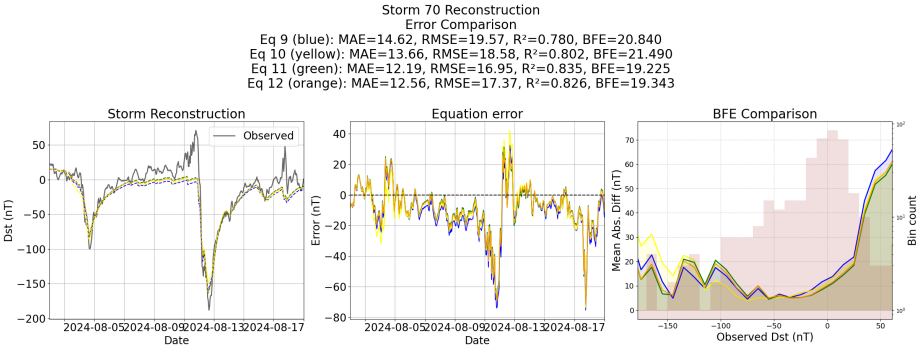

Figure 37: Reconstruction of storm 70 using the Equations generated from the unconstrained symbolic regression with the derived features

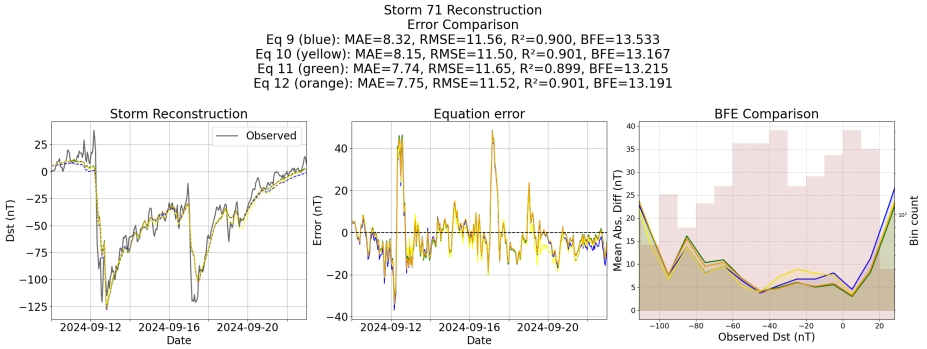

Figure 38: Reconstruction of storm 71 using the Equations generated from the unconstrained symbolic regression with the derived features

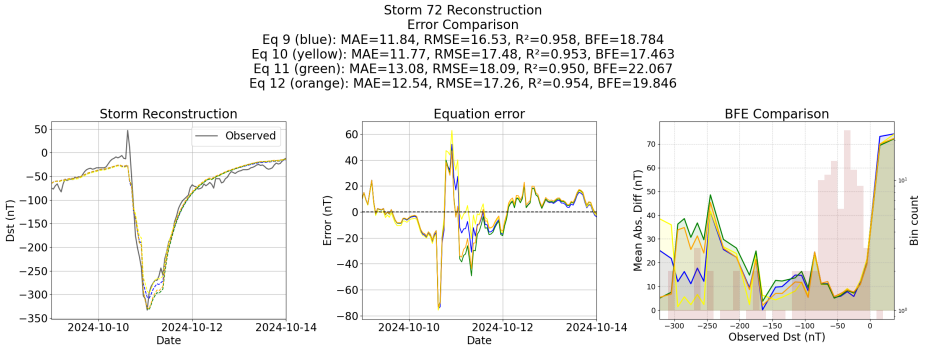

Figure 39: Reconstruction of storm 72 using the Equations generated from the unconstrained symbolic regression with the derived features

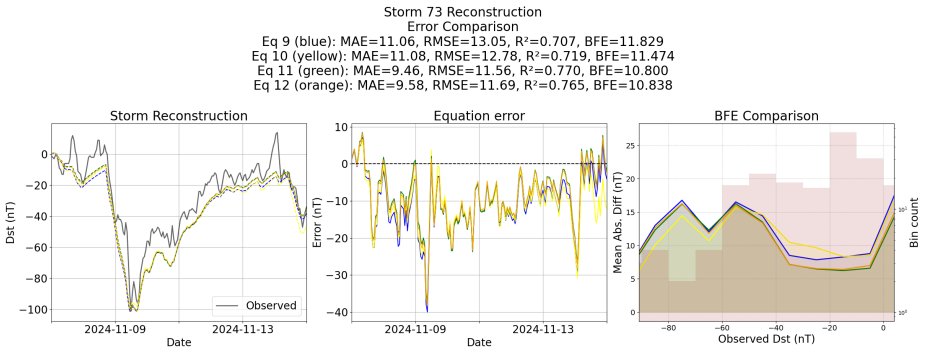

Figure 40: Reconstruction of storm 73 using the Equations generated from the unconstrained symbolic regression with the derived features

3 Templated primitive equations

3.1 Equations

$$\frac{dDst}{dt} = -0.00199224768611389(B_s + 2.152752)[B_s(N - 4.7333016) + V] + (0.015329162Dst - 1.4046433)^2 \quad (13)$$

$$\frac{dDst}{dt} = -0.0020573507(B_s + 1.775414)[B_s(N - \sqrt{B_s}) + V] + (0.015831897Dst - 1.3089455)^2 \quad (14)$$

$$\frac{dDst}{dt} = -0.0018313164(B_s + 2.1569605)(B_sN + V) + (0.015403083Dst - 1.3434031)^2 \quad (15)$$

$$\frac{dDst}{dt} = 0.0053468808N - 0.0053468808V[B_s(3.5000103 \times 10^{-5}NV + 0.301232) + 0.55653155] + (0.015961226Dst - 1.2015269)^2 \quad (16)$$

$$\frac{dDst}{dt} = -0.0021015273(B_s + 1.6439178)[B_s(N - \sqrt{B}) + V] + (0.015890777Dst - 1.2798462)^2 \quad (17)$$

$$\frac{dDst}{dt} = -0.0018150203(B_s + 2.0452397)(B_sN + V) + (0.015376654Dst - 1.3081883)^2 \quad (18)$$

$$\frac{dDst}{dt} = -0.0016007437V(0.00011782655B_sNV + B_s + 1.3935444) + (0.016486706Dst - 1.0748392)^2 \quad (19)$$

3.2 Metrics per equation per storm

Table 3: RMSE, MAE, R^2, and BFE for the templated primitive-feature equations (Eqs. 13–15) across the 20 test storms.

Storm	Equation 13				Equation 14				Equation 15			
	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE
54	9.746	6.839	0.766	12.618	9.170	6.307	0.792	12.136	9.856	6.843	0.760	13.042
55	15.457	12.125	0.803	17.293	15.477	12.047	0.802	16.816	15.441	12.018	0.803	16.969
56	11.924	9.441	0.688	13.777	11.763	9.264	0.696	13.862	11.754	9.165	0.697	13.684
57	10.549	8.067	0.717	10.448	10.641	8.014	0.712	9.156	10.215	7.755	0.735	9.319
58	9.316	6.358	0.790	15.774	8.454	5.979	0.827	14.344	8.749	6.008	0.815	14.974
59	19.212	16.201	0.628	12.302	18.932	16.047	0.639	11.733	18.954	16.054	0.638	12.257
60	16.652	13.110	0.801	20.074	16.514	13.513	0.805	18.459	15.869	13.114	0.820	17.448
61	11.790	8.401	0.917	17.170	11.884	8.887	0.916	17.416	12.329	8.682	0.909	18.069
62	12.659	10.555	0.627	14.579	12.744	10.837	0.622	13.081	12.294	10.326	0.649	13.216
63	13.332	8.724	0.871	20.154	12.608	8.299	0.885	18.443	12.574	8.374	0.885	18.675
64	10.038	7.676	0.806	13.468	9.427	7.313	0.829	12.233	9.387	7.269	0.830	12.462
65	13.401	10.465	0.676	13.184	14.197	10.716	0.637	12.369	13.374	10.216	0.678	12.456
66	15.870	11.576	0.654	23.129	16.017	11.755	0.648	22.630	15.824	11.495	0.656	22.954
67	9.192	6.766	0.850	14.298	8.377	6.240	0.875	11.865	8.462	6.287	0.873	12.158
68	18.717	15.040	0.976	17.226	19.614	15.943	0.973	17.929	19.305	15.507	0.974	18.261
69	10.618	8.059	0.802	12.012	10.488	7.994	0.807	11.910	10.293	7.840	0.814	11.671
70	17.531	12.099	0.823	22.030	17.811	12.353	0.818	20.655	17.641	12.166	0.821	20.723
71	12.316	7.995	0.887	14.043	12.228	8.468	0.888	14.190	12.279	8.319	0.887	14.250
72	24.357	13.963	0.909	26.333	23.984	14.017	0.912	25.496	25.272	15.369	0.902	30.512
73	7.782	5.660	0.896	6.938	9.818	7.371	0.834	9.094	8.737	6.441	0.868	7.930
Mean	13.523	9.956	0.794	15.843	13.507	10.068	0.796	15.191	13.431	9.962	0.801	15.552
Global	13.841	9.805	0.888	25.278	13.824	9.903	0.889	24.735	13.749	9.775	0.890	25.717

Table 4: RMSE, MAE, R^2, and BFE for the remaining templated primitive-feature equations (Eqs. 16–19) across the 20 test storms.

Storm	Equation 16				Equation 17				Equation 18				Equation 19			
	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE
54	8.695	6.595	0.813	11.029	9.039	6.153	0.798	12.028	9.951	6.892	0.756	13.239	8.674	6.589	0.814	11.302
55	16.373	12.792	0.778	18.150	15.514	12.034	0.801	16.761	15.517	12.059	0.801	17.140	16.449	12.927	0.776	18.165
56	12.344	9.641	0.666	15.320	11.675	9.185	0.701	13.812	11.596	9.034	0.705	13.571	11.832	9.306	0.693	14.710
57	10.053	7.852	0.743	10.416	10.756	8.088	0.706	9.050	10.129	7.717	0.739	9.454	9.834	7.633	0.754	9.969
58	8.594	5.818	0.822	14.660	8.416	5.991	0.829	14.242	8.737	5.973	0.816	15.027	8.344	5.728	0.832	14.176
59	16.777	14.235	0.717	10.676	18.906	16.055	0.640	11.683	18.756	15.919	0.646	12.318	15.742	13.451	0.750	10.586
60	15.882	12.109	0.819	20.218	16.600	13.636	0.803	18.423	15.906	13.180	0.819	17.604	15.842	12.059	0.820	20.299
61	10.560	7.590	0.934	16.165	11.791	8.984	0.917	17.053	12.411	8.672	0.908	18.248	10.534	7.737	0.934	16.059
62	11.927	9.866	0.669	13.424	12.822	10.947	0.618	12.960	12.240	10.293	0.652	13.171	11.981	9.956	0.666	12.976
63	11.698	7.691	0.901	17.293	12.693	8.308	0.883	18.477	12.509	8.251	0.886	18.606	11.964	8.226	0.896	16.950
64	9.156	7.029	0.839	11.212	9.368	7.271	0.831	12.103	9.315	7.212	0.833	12.457	9.087	6.965	0.841	10.964
65	13.843	10.913	0.655	16.027	14.477	10.818	0.622	12.146	13.390	10.162	0.677	12.437	13.969	10.625	0.648	14.932
66	15.174	11.711	0.684	19.782	16.134	11.870	0.643	22.638	16.001	11.648	0.649	23.151	15.964	12.487	0.650	20.136
67	9.310	7.043	0.846	13.905	8.365	6.286	0.876	11.544	8.550	6.332	0.870	12.204	9.319	7.000	0.846	13.461
68	33.211	23.966	0.924	38.076	19.717	16.059	0.973	18.136	19.174	15.366	0.975	18.115	32.338	23.756	0.927	36.142
69	10.758	8.190	0.797	12.280	10.510	8.004	0.806	11.909	10.256	7.795	0.815	11.613	10.819	8.198	0.794	12.287
70	17.138	11.729	0.831	18.883	17.837	12.408	0.817	20.401	17.664	12.197	0.821	20.835	17.587	12.131	0.822	18.928
71	11.696	7.879	0.898	13.638	12.371	8.680	0.886	14.359	12.392	8.422	0.885	14.351	11.721	8.024	0.897	13.734
72	20.032	13.808	0.938	25.985	24.028	14.120	0.911	25.515	25.501	15.495	0.900	30.818	19.380	13.643	0.942	24.558
73	9.619	7.174	0.841	8.946	10.249	7.843	0.819	9.773	8.860	6.610	0.865	8.114	10.847	8.388	0.797	10.438
Mean	13.642	10.182	0.806	16.304	13.564	10.137	0.794	15.151	13.443	9.961	0.801	15.624	13.611	10.241	0.805	16.039
Global	13.953	9.787	0.887	33.938	13.877	9.969	0.888	24.747	13.761	9.775	0.890	25.720	13.927	9.867	0.887	32.250

3.3 Figures per equation per storm

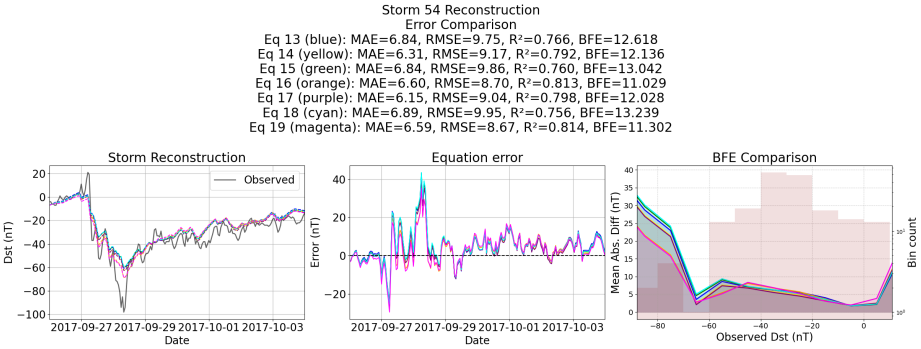

Figure 41: Reconstruction of storm 54 using the Equations generated from the templated symbolic regression with the primitive features

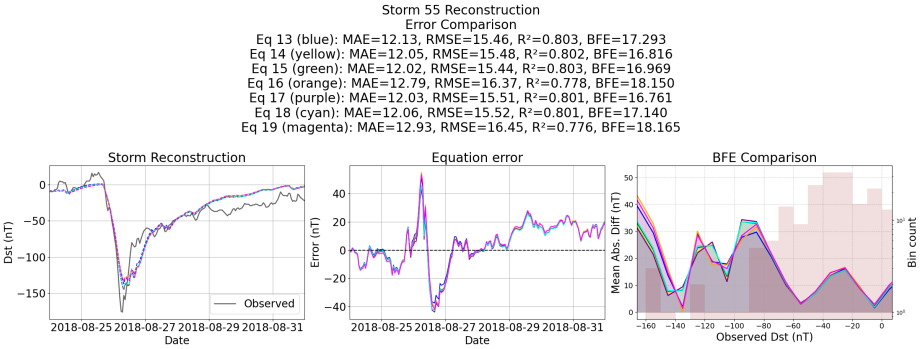

Figure 42: Reconstruction of storm 55 using the Equations generated from the templated symbolic regression with the primitive features

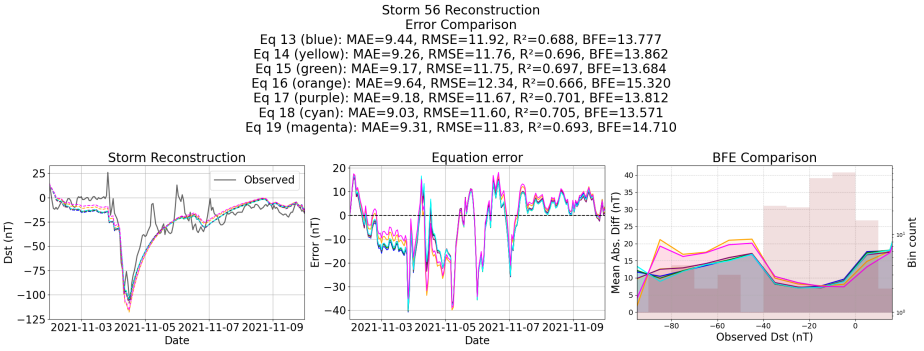

Figure 43: Reconstruction of storm 56 using the Equations generated from the templated symbolic regression with the primitive features

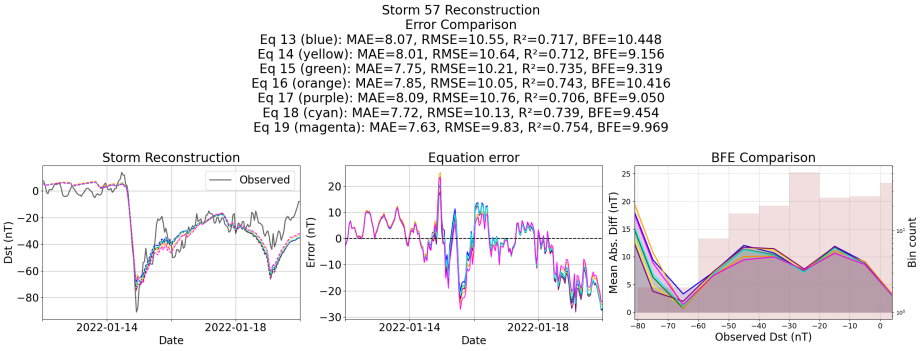

Figure 44: Reconstruction of storm 57 using the Equations generated from the templated symbolic regression with the primitive features

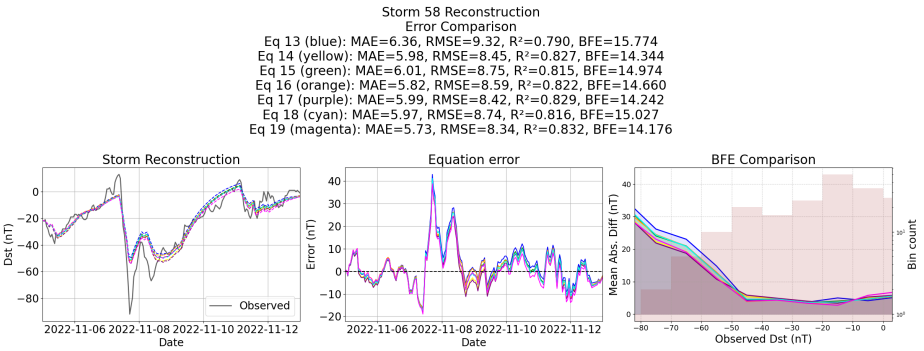

Figure 45: Reconstruction of storm 58 using the Equations generated from the templated symbolic regression with the primitive features

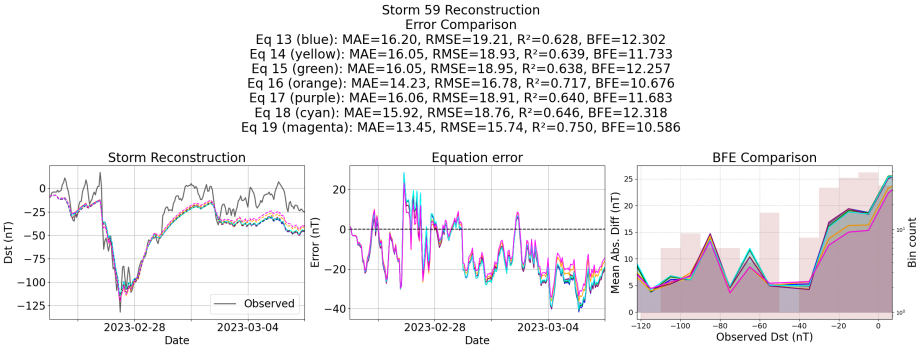

Figure 46: Reconstruction of storm 59 using the Equations generated from the templated symbolic regression with the primitive features

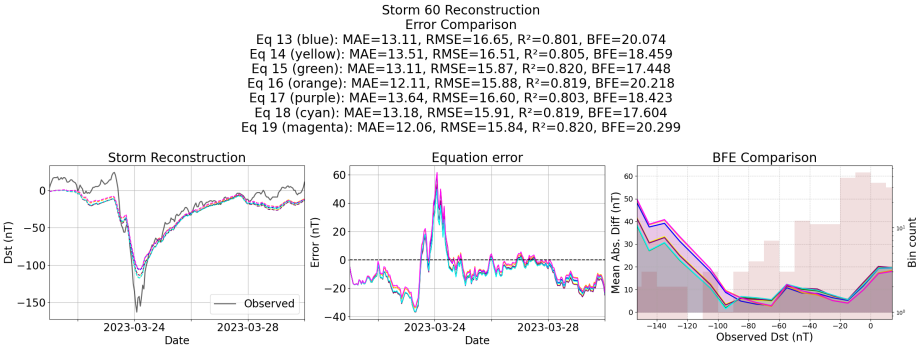

Figure 47: Reconstruction of storm 60 using the Equations generated from the templated symbolic regression with the primitive features

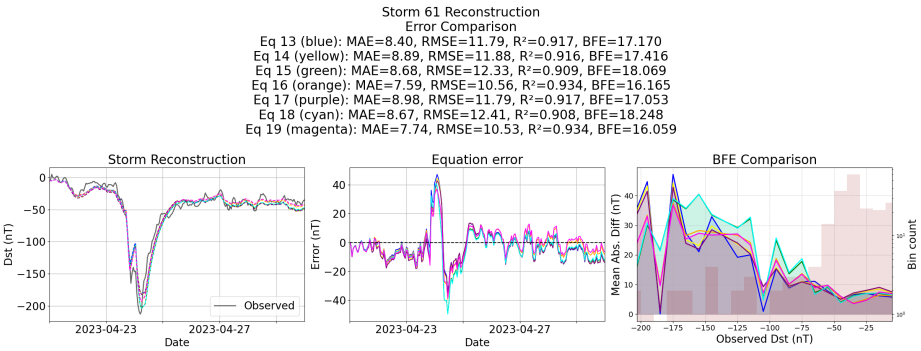

Figure 48: Reconstruction of storm 61 using the Equations generated from the templated symbolic regression with the primitive features

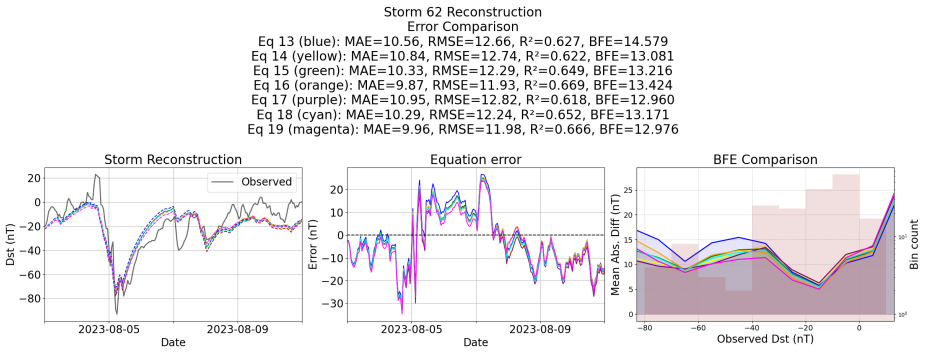

Figure 49: Reconstruction of storm 62 using the Equations generated from the templated symbolic regression with the primitive features

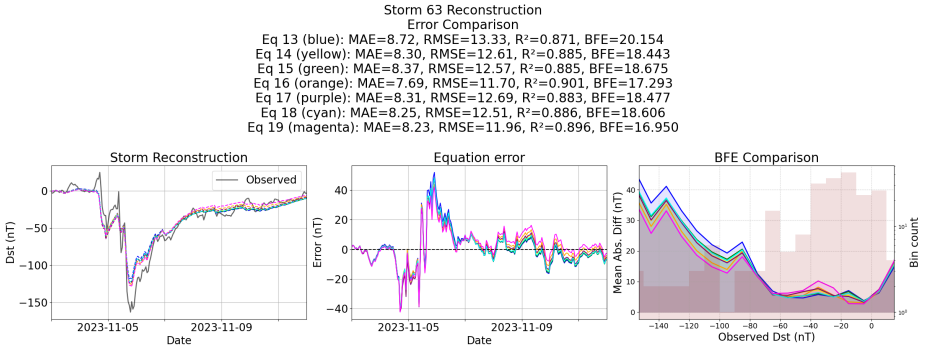

Figure 50: Reconstruction of storm 63 using the Equations generated from the templated symbolic regression with the primitive features

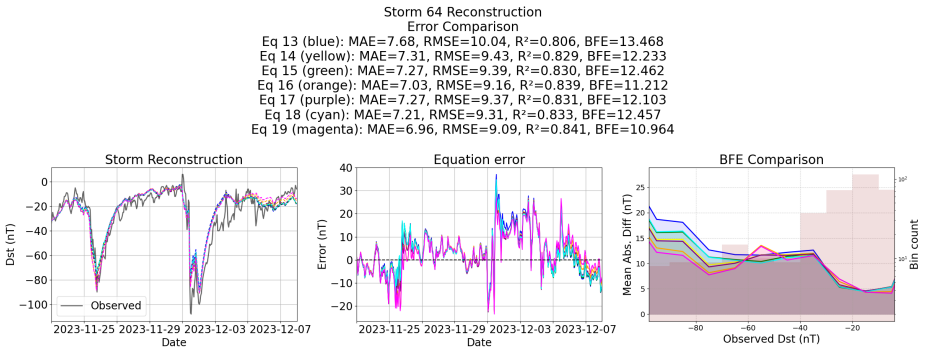

Figure 51: Reconstruction of storm 64 using the Equations generated from the templated symbolic regression with the primitive features

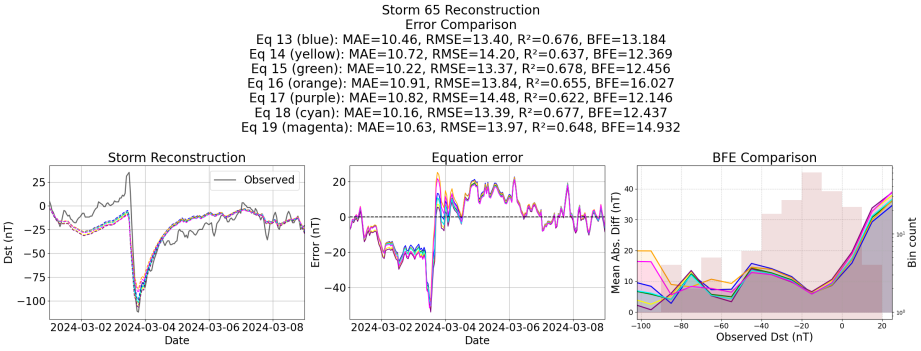

Figure 52: Reconstruction of storm 65 using the Equations generated from the templated symbolic regression with the primitive features

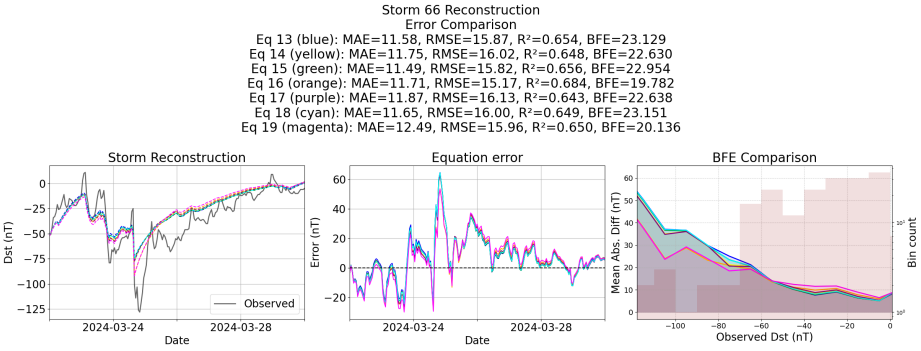

Figure 53: Reconstruction of storm 66 using the Equations generated from the templated symbolic regression with the primitive features

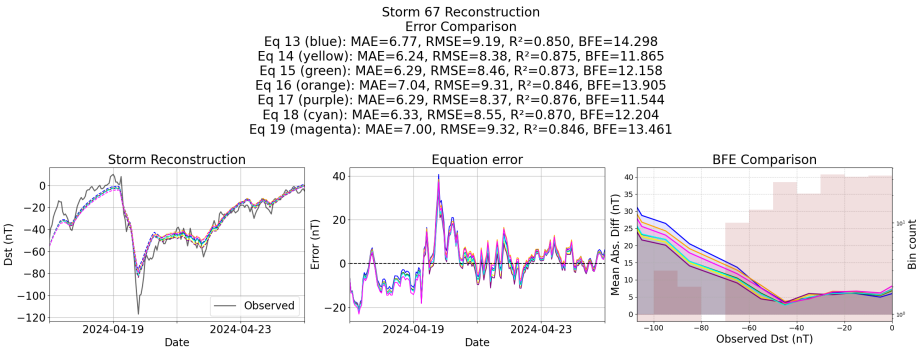

Figure 54: Reconstruction of storm 67 using the Equations generated from the templated symbolic regression with the primitive features

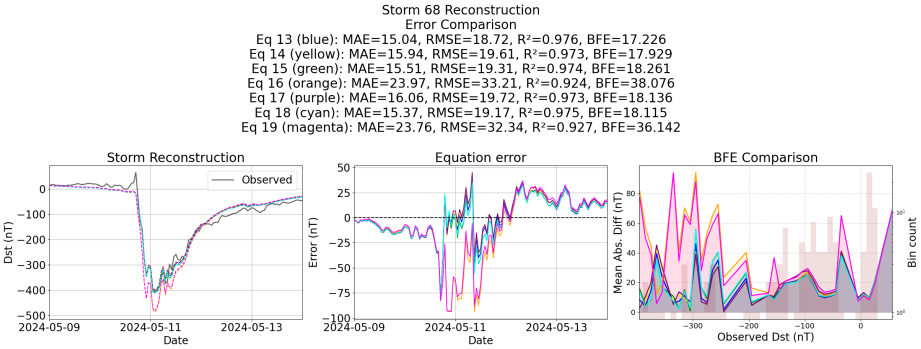

Figure 55: Reconstruction of storm 68 using the Equations generated from the templated symbolic regression with the primitive features

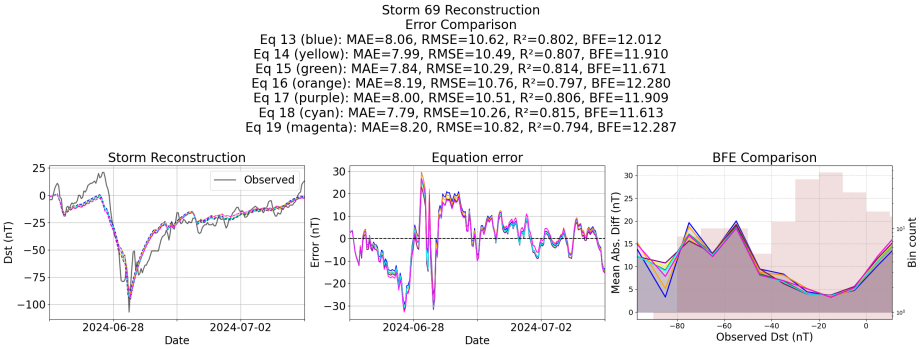

Figure 56: Reconstruction of storm 69 using the Equations generated from the templated symbolic regression with the primitive features

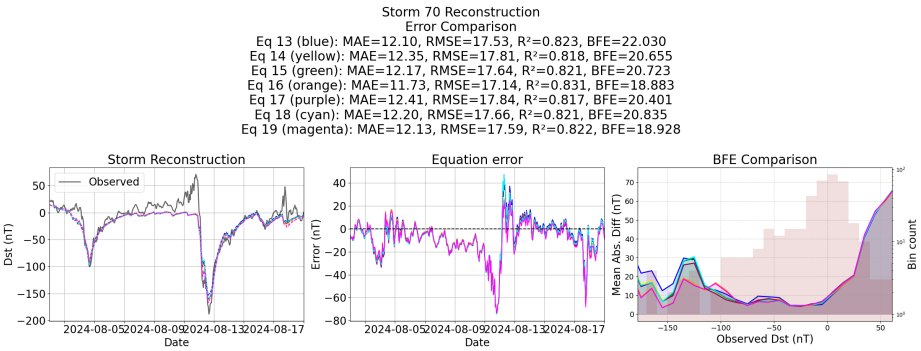

Figure 57: Reconstruction of storm 70 using the Equations generated from the templated symbolic regression with the primitive features

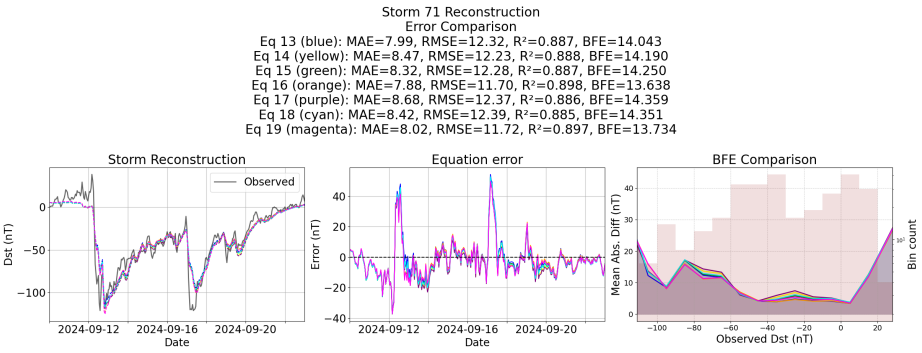

Figure 58: Reconstruction of storm 71 using the Equations generated from the templated symbolic regression with the primitive features

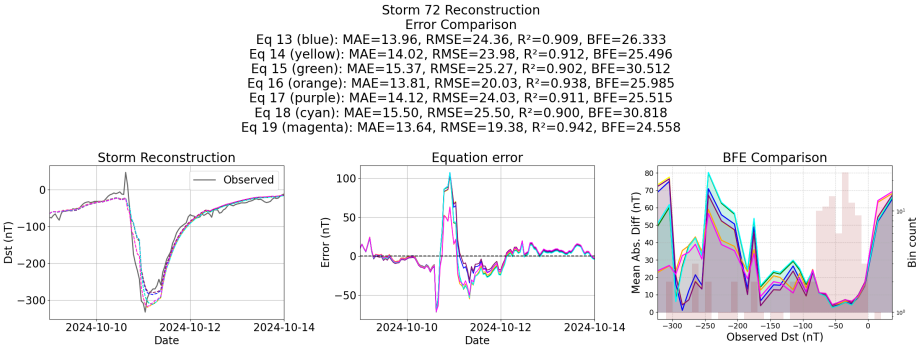

Figure 59: Reconstruction of storm 72 using the Equations generated from the templated symbolic regression with the primitive features

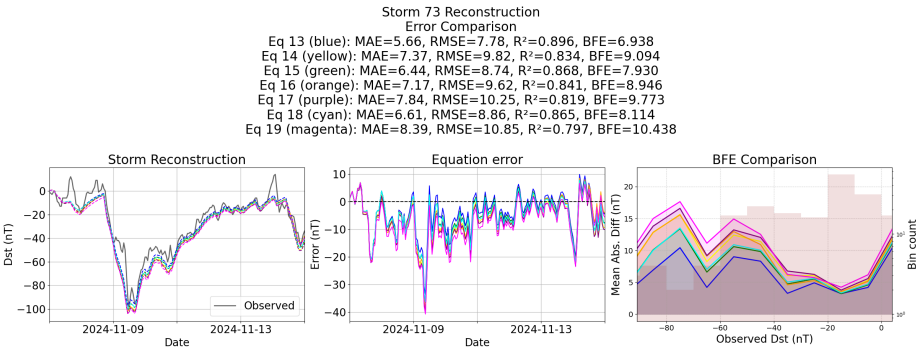

Figure 60: Reconstruction of storm 73 using the Equations generated from the templated symbolic regression with the primitive features

4 Templated derived equations

4.1 Equations

$$\frac{dDst}{dt} = -0.0010713526VB_s\sqrt{P_{dyn} + 1.32442} + (0.01838707Dst - 0.5912093)^2 \tag{21}$$

$$\frac{dDst}{dt} = -0.0010842164VB_s\sqrt{P_{dyn} + 1.1162246} + (0.01812232Dst - 0.5913376)^2 \tag{22}$$

$$\frac{dDst}{dt} = -0.0010888062VB_s\sqrt{P_{dyn} + 1.1051167} + (0.018209398Dst - 0.59240085)^2 \tag{23}$$

4.2 Metrics per equation per storm

Table 5: RMSE, MAE, R^2, and BFE for the templated derived-feature equations (Eqs. 21–23) across the 20 test storms.

Storm	Equation 21				Equation 22				Equation 23			
	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE
54	9.082	7.150	0.796	11.621	9.134	7.233	0.794	11.626	9.192	7.301	0.791	11.646
55	17.429	13.783	0.749	19.559	17.565	13.841	0.745	19.785	17.584	13.887	0.744	19.750
56	13.276	10.963	0.613	15.586	13.524	11.248	0.599	15.826	13.529	11.280	0.598	15.788
57	9.580	7.666	0.767	12.125	9.579	7.657	0.767	12.267	9.543	7.634	0.768	12.215
58	9.166	6.390	0.797	15.585	9.215	6.349	0.795	15.792	9.192	6.313	0.796	15.777
59	12.207	10.241	0.850	9.432	12.083	10.089	0.853	9.452	11.975	9.989	0.856	9.410
60	17.618	11.829	0.778	26.808	18.391	11.915	0.758	28.592	18.421	11.915	0.757	28.684
61	10.680	8.252	0.932	14.553	10.784	8.343	0.931	14.539	10.824	8.392	0.930	14.458
62	11.395	9.123	0.698	12.821	11.196	8.935	0.709	13.002	11.178	8.931	0.710	13.034
63	14.976	12.168	0.837	18.193	14.829	12.015	0.840	17.878	14.928	12.134	0.838	17.997
64	10.131	8.022	0.802	11.184	10.153	8.046	0.802	11.235	10.221	8.115	0.799	11.288
65	14.250	10.391	0.634	16.795	13.929	10.181	0.650	16.666	13.930	10.221	0.650	16.694
66	18.264	15.417	0.542	20.121	18.151	15.388	0.548	19.934	18.236	15.467	0.544	20.018
67	11.029	8.252	0.784	15.548	11.425	8.652	0.768	16.418	11.440	8.702	0.767	16.491
68	27.918	20.659	0.946	30.007	28.983	20.997	0.942	31.809	28.990	21.060	0.942	31.609
69	10.958	8.250	0.789	12.099	10.985	8.318	0.788	12.133	11.026	8.381	0.786	12.184
70	16.847	12.101	0.837	19.179	16.758	12.020	0.839	19.217	16.704	11.946	0.840	19.175
71	11.688	7.816	0.898	13.279	11.733	7.723	0.897	13.228	11.686	7.649	0.898	13.159
72	18.315	13.270	0.949	22.361	18.600	13.338	0.947	23.274	18.413	13.266	0.948	22.839
73	11.560	9.467	0.770	10.805	11.490	9.391	0.772	10.771	11.369	9.265	0.777	10.648
Mean	13.818	10.560	0.788	16.383	13.925	10.584	0.787	16.672	13.919	10.592	0.787	16.643
Global	14.013	10.250	0.886	28.791	14.123	10.258	0.884	30.130	14.119	10.263	0.884	29.894

4.3 Figures per storm

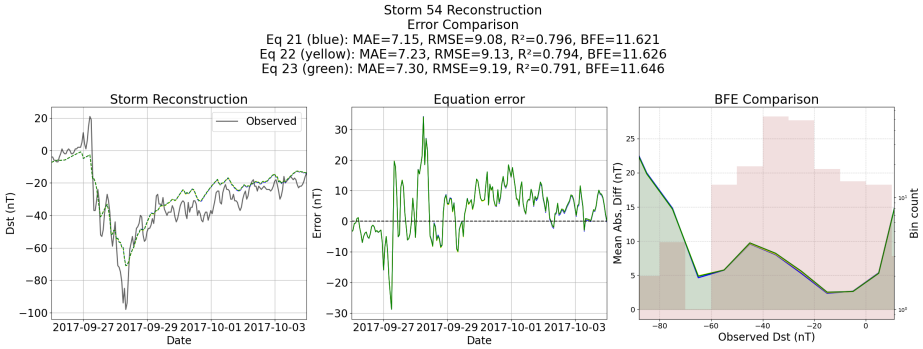

Figure 61: Reconstruction of storm 54 using the Equations generated from the templated symbolic regression with the derived features

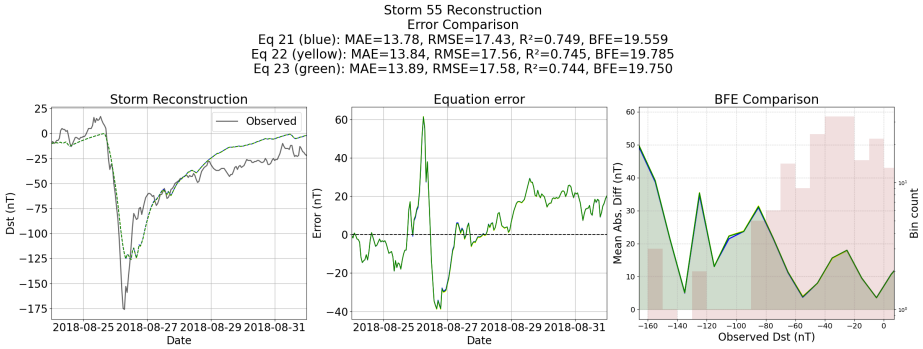

Figure 62: Reconstruction of storm 55 using the Equations generated from the templated symbolic regression with the derived features

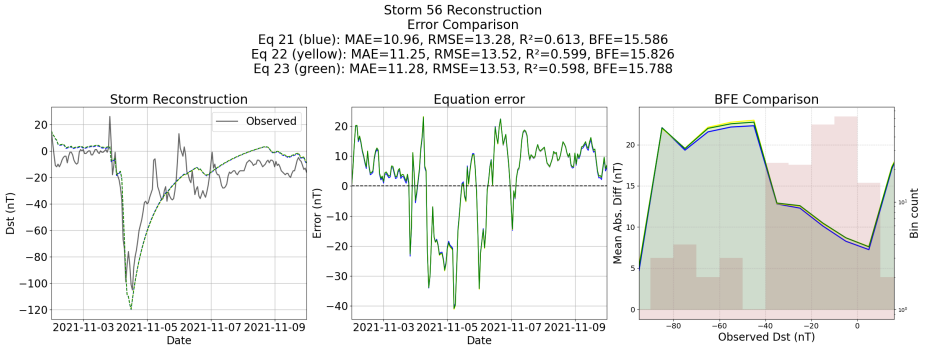

Figure 63: Reconstruction of storm 56 using the Equations generated from the templated symbolic regression with the derived features

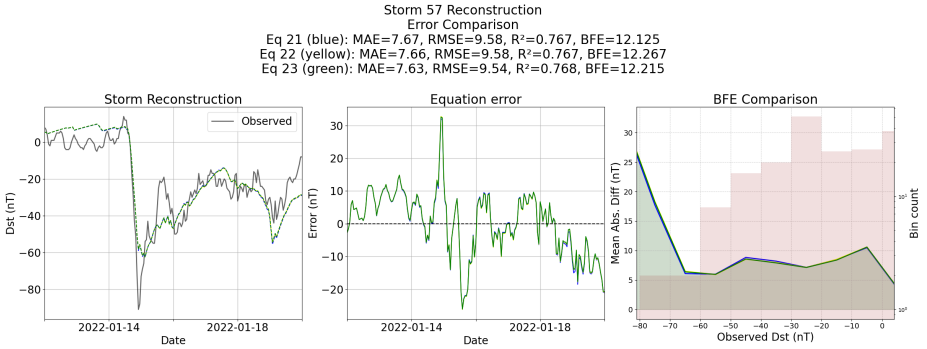

Figure 64: Reconstruction of storm 57 using the Equations generated from the templated symbolic regression with the derived features

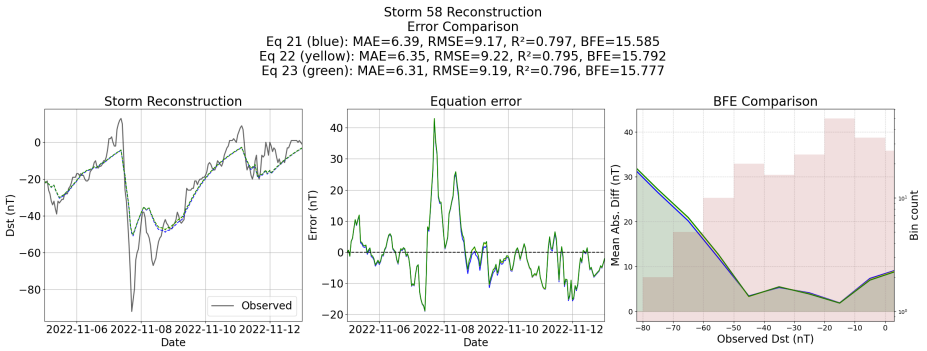

Figure 65: Reconstruction of storm 58 using the Equations generated from the templated symbolic regression with the derived features

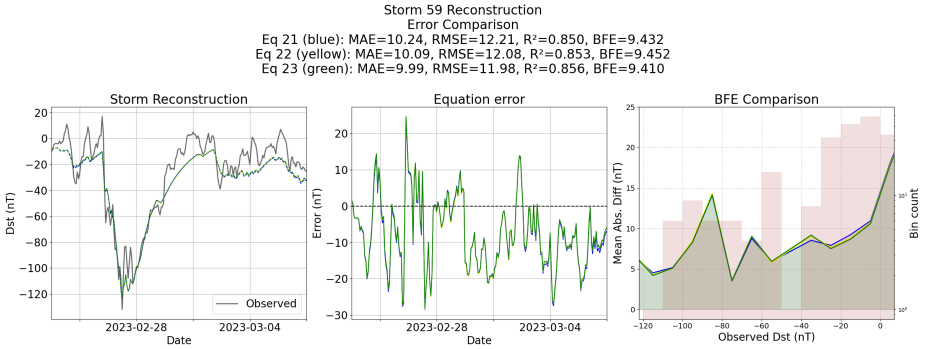

Figure 66: Reconstruction of storm 59 using the Equations generated from the templated symbolic regression with the derived features

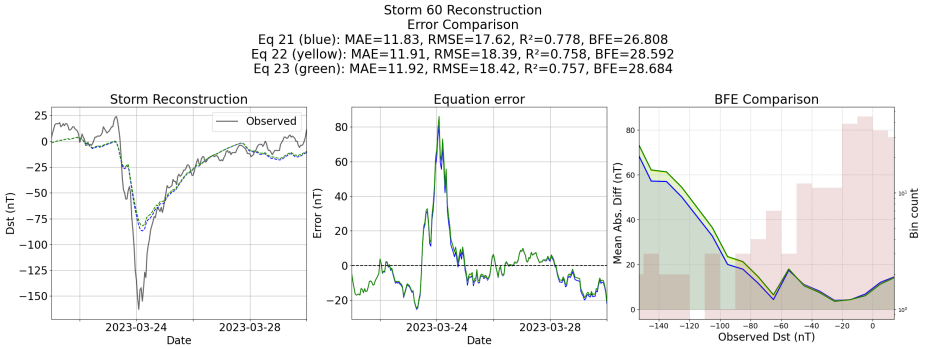

Figure 67: Reconstruction of storm 60 using the Equations generated from the templated symbolic regression with the derived features

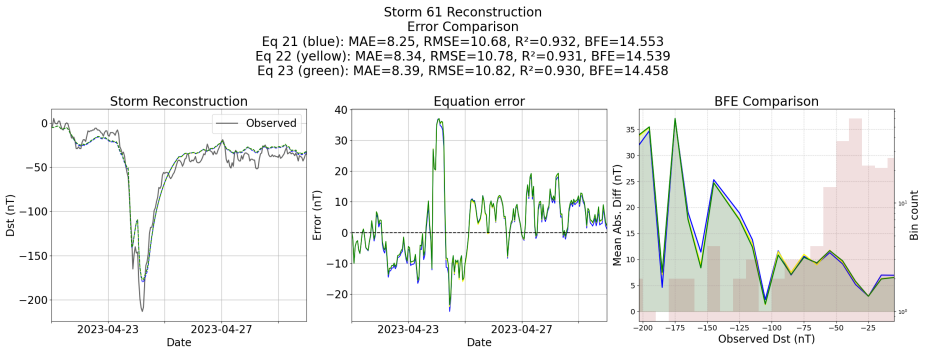

Figure 68: Reconstruction of storm 61 using the Equations generated from the templated symbolic regression with the derived features

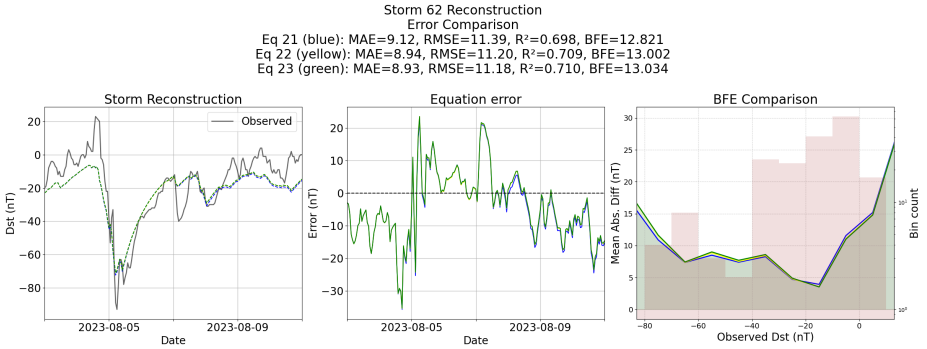

Figure 69: Reconstruction of storm 62 using the Equations generated from the templated symbolic regression with the derived features

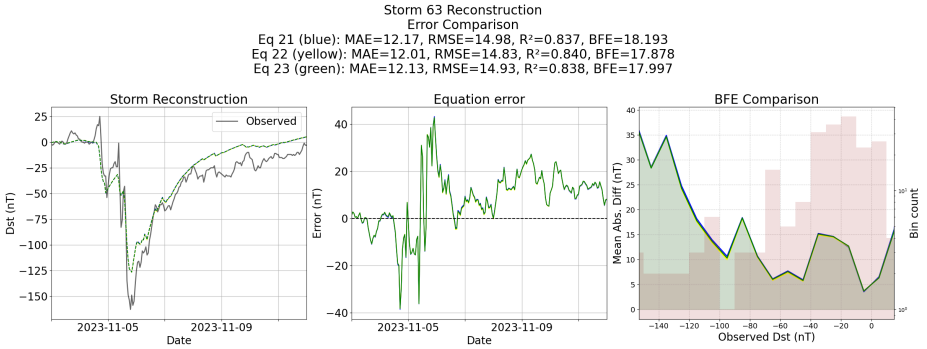

Figure 70: Reconstruction of storm 63 using the Equations generated from the templated symbolic regression with the derived features

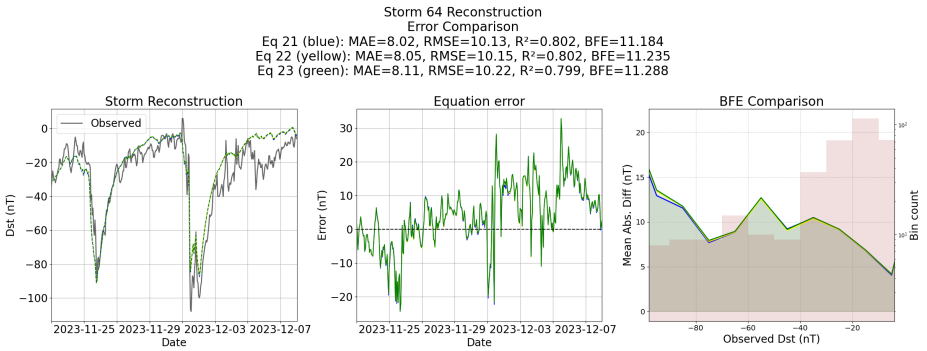

Figure 71: Reconstruction of storm 64 using the Equations generated from the templated symbolic regression with the derived features

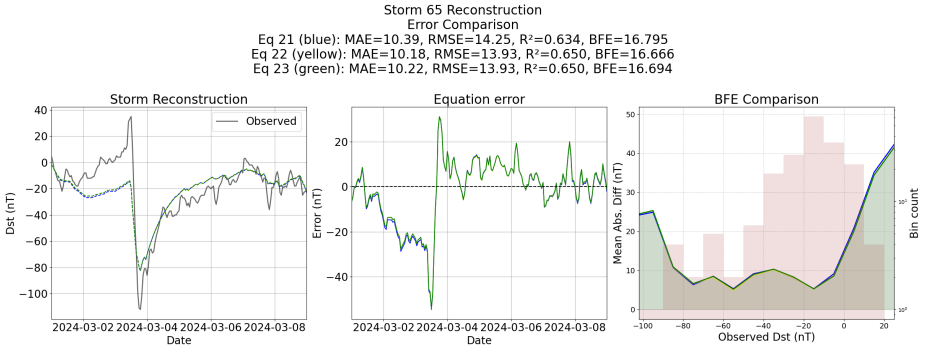

Figure 72: Reconstruction of storm 65 using the Equations generated from the templated symbolic regression with the derived features

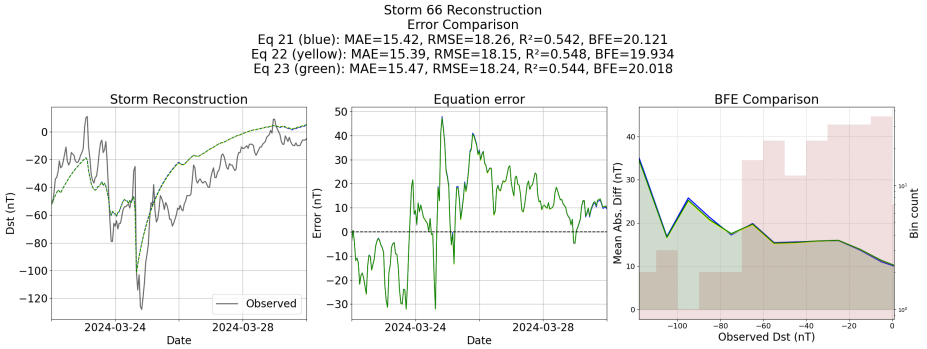

Figure 73: Reconstruction of storm 66 using the Equations generated from the templated symbolic regression with the derived features

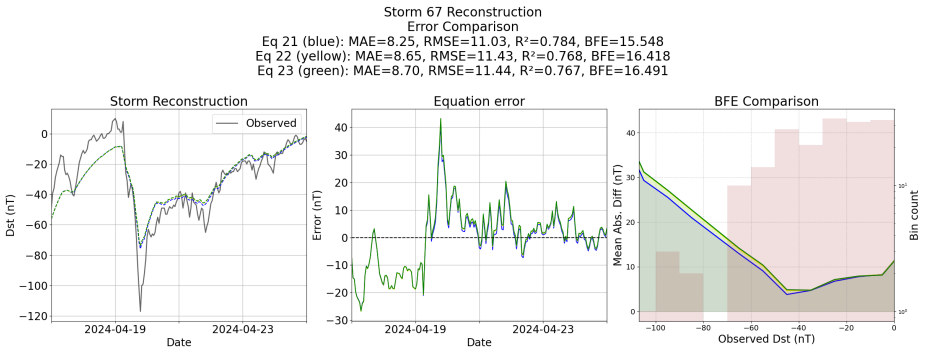

Figure 74: Reconstruction of storm 67 using the Equations generated from the templated symbolic regression with the derived features

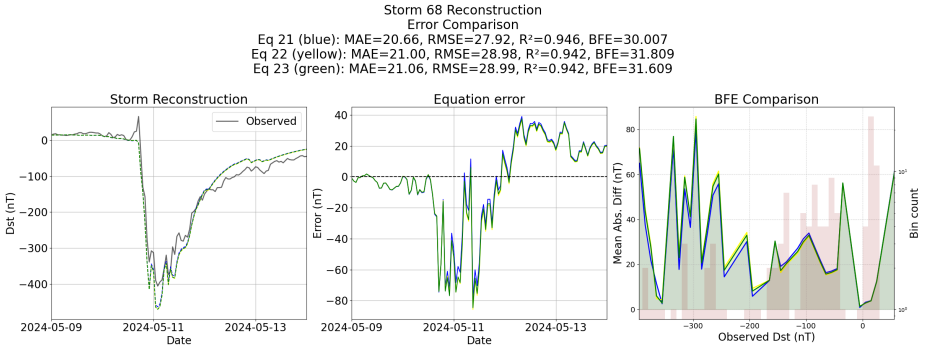

Figure 75: Reconstruction of storm 68 using the Equations generated from the templated symbolic regression with the derived features

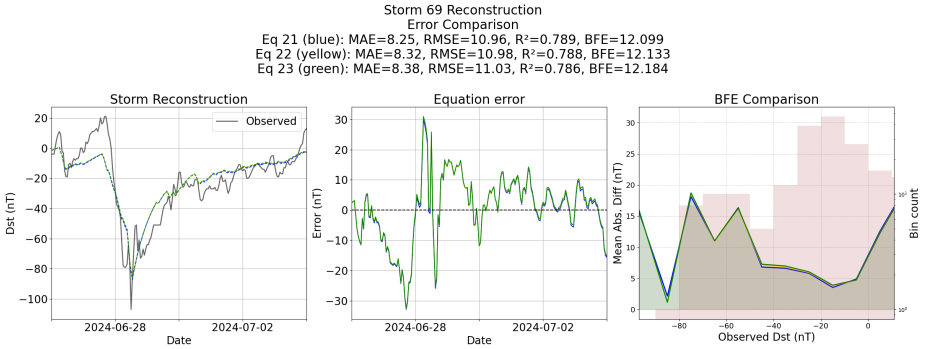

Figure 76: Reconstruction of storm 69 using the Equations generated from the templated symbolic regression with the derived features

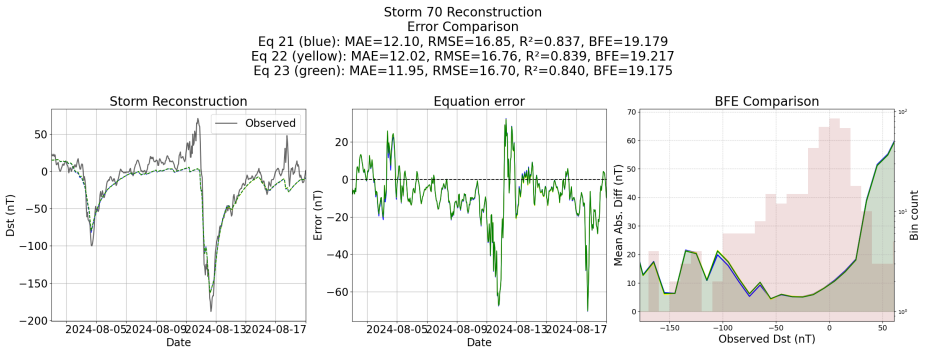

Figure 77: Reconstruction of storm 70 using the Equations generated from the templated symbolic regression with the derived features

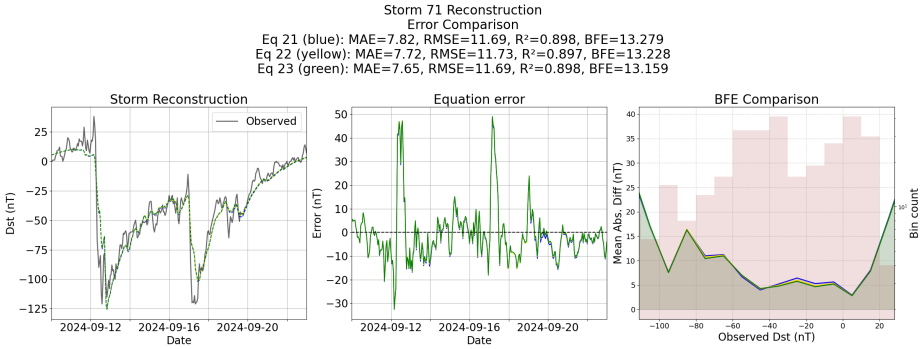

Figure 78: Reconstruction of storm 71 using the Equations generated from the templated symbolic regression with the derived features

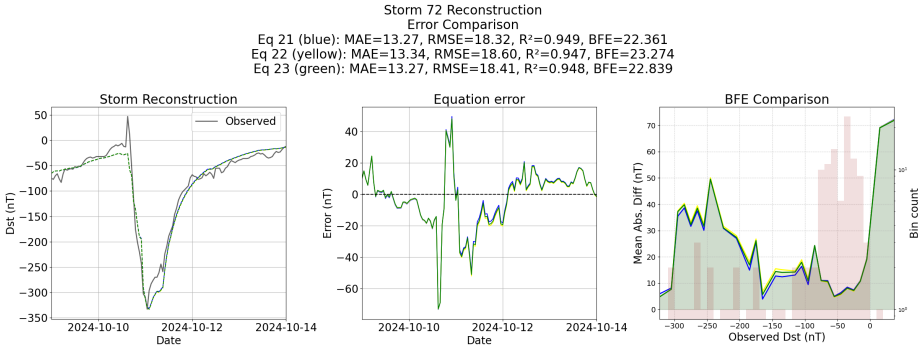

Figure 79: Reconstruction of storm 72 using the Equations generated from the templated symbolic regression with the derived features

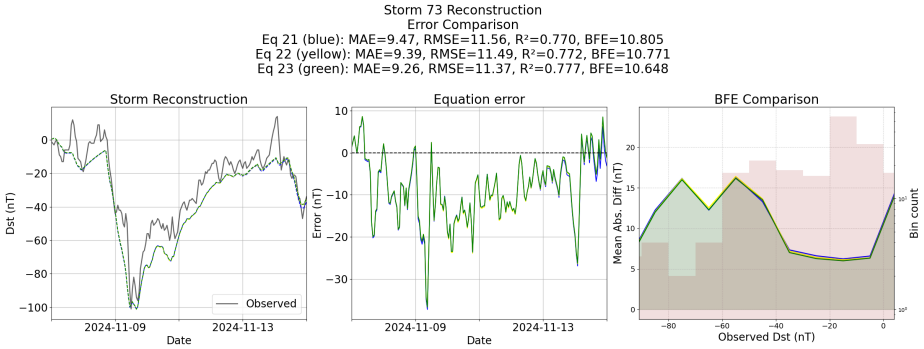

Figure 80: Reconstruction of storm 73 using the Equations generated from the templated symbolic regression with the derived features

5 Comparison against baselines

5.1 Baseline metrics per storm

Table 6: RMSE, MAE, R^2, and BFE for the five baseline models across the 20 test storms.

Storm	Burton				OBM				DDM#1				DDM#2				DDM#3			
	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE	RMSE	MAE	R^2	BFE
54	22.374	19.269	-0.235	22.860	12.935	10.340	0.587	15.495	17.685	15.097	0.228	21.130	17.034	14.718	0.284	19.285	17.429	14.902	0.250	20.945
55	22.579	19.589	0.579	19.866	17.163	14.453	0.757	17.013	21.696	17.787	0.611	23.435	20.995	17.673	0.636	22.088	22.443	18.174	0.584	25.386
56	14.342	9.960	0.549	20.707	13.463	9.386	0.602	17.920	13.653	10.595	0.591	13.923	11.711	9.330	0.699	13.104	11.379	9.160	0.716	12.828
57	19.247	14.483	0.058	15.794	10.407	8.641	0.725	10.567	8.769	6.755	0.805	13.120	8.303	6.470	0.825	11.442	8.671	6.332	0.809	13.884
58	9.601	7.210	0.777	11.978	10.385	8.599	0.739	13.090	12.246	9.298	0.638	20.109	11.990	9.587	0.653	19.032	11.863	8.935	0.660	19.971
59	16.064	12.179	0.740	16.352	16.381	14.167	0.730	10.468	11.365	9.089	0.870	14.983	10.840	8.734	0.882	13.497	12.355	9.579	0.846	16.760
60	25.566	20.035	0.532	33.653	19.034	16.422	0.741	15.765	22.099	14.076	0.650	36.468	13.320	10.330	0.873	17.937	14.717	11.742	0.845	19.773
61	40.570	21.186	0.019	59.661	18.231	13.010	0.802	24.712	21.954	16.492	0.713	28.912	22.490	16.827	0.698	30.617	20.469	15.215	0.750	30.187
62	14.360	10.614	0.521	13.397	16.053	13.638	0.401	13.831	10.150	7.609	0.760	14.844	10.406	7.936	0.748	14.058	11.805	9.353	0.676	16.882
63	22.708	18.801	0.625	24.688	16.108	12.907	0.811	21.266	21.097	16.844	0.677	29.948	20.014	16.071	0.709	28.117	22.085	17.377	0.646	32.083
64	15.235	12.079	0.553	18.002	8.640	6.722	0.856	10.293	11.924	9.683	0.726	14.891	11.958	9.921	0.725	15.055	13.759	11.130	0.635	18.588
65	15.580	13.082	0.562	20.237	14.909	11.609	0.599	17.679	14.535	11.156	0.619	22.931	14.063	10.704	0.643	22.233	16.593	12.906	0.504	26.521
66	29.270	20.808	-0.176	33.470	17.886	13.597	0.561	21.849	20.760	17.779	0.409	26.472	22.099	17.823	0.330	27.988	19.082	15.244	0.500	25.992
67	11.747	8.490	0.755	11.658	12.715	10.695	0.713	9.476	16.003	13.857	0.545	22.933	14.922	12.664	0.604	18.323	13.803	11.556	0.661	19.324
68	72.495	53.764	0.636	76.973	44.485	32.605	0.863	50.547	26.232	19.685	0.952	29.756	41.573	29.791	0.880	48.472	50.339	34.755	0.824	54.647
69	16.115	11.370	0.544	15.048	11.346	8.321	0.774	11.393	14.239	11.912	0.644	17.502	13.769	11.665	0.667	15.952	12.629	10.200	0.720	15.432
70	21.401	17.667	0.737	25.743	21.488	17.172	0.734	22.645	20.124	13.351	0.767	27.058	19.950	13.819	0.771	24.837	22.293	15.003	0.714	27.983
71	17.638	14.075	0.768	17.069	15.199	12.909	0.828	16.584	15.912	11.077	0.811	17.363	16.262	12.396	0.803	18.055	17.329	12.601	0.776	19.930
72	98.544	49.327	-0.490	117.002	27.055	17.113	0.888	31.379	64.699	51.497	0.358	69.502	53.242	37.024	0.565	66.197	44.726	30.412	0.693	55.474
73	17.435	12.718	0.476	14.427	14.321	10.334	0.647	10.115	9.766	8.073	0.836	12.595	9.024	7.395	0.860	10.469	9.515	7.435	0.844	12.582
Mean	26.144	18.335	0.426	29.429	16.910	13.132	0.718	18.104	18.745	14.586	0.660	23.894	18.198	14.044	0.693	22.838	18.664	14.101	0.683	24.259
Global	28.711	16.825	0.520	78.078	17.337	12.709	0.825	44.103	19.864	13.524	0.770	38.888	19.055	13.157	0.789	45.264	19.546	13.358	0.778	46.379

5.2 Figures per storm

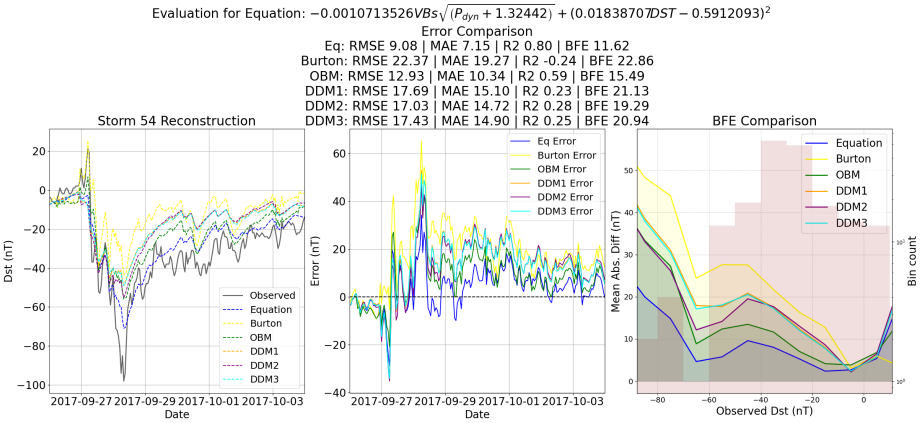

Figure 81: Reconstruction of storm 54 using Eq. 21 against the five baseline models

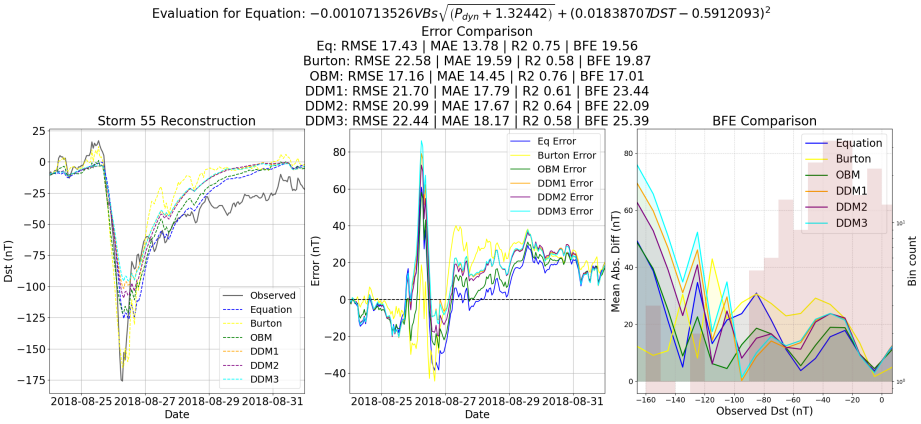

Figure 82: Reconstruction of storm 55 using Eq. 21 against the five baseline models

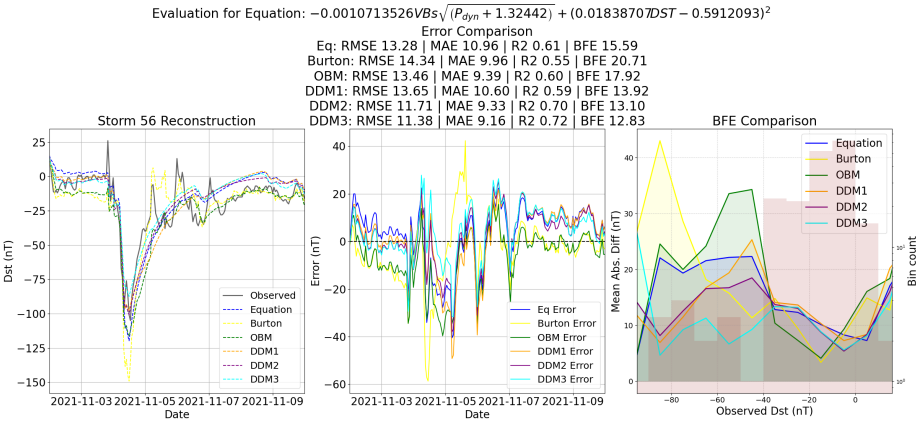

Figure 83: Reconstruction of storm 56 using Eq. 21 against the five baseline models

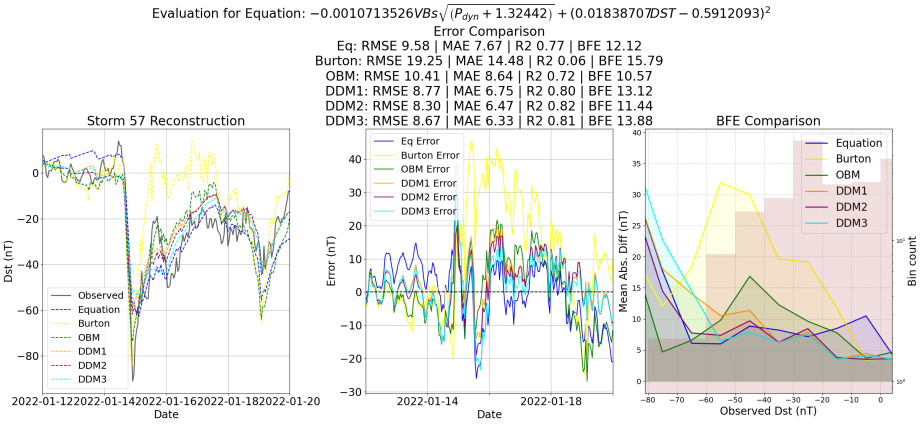

Figure 84: Reconstruction of storm 57 using Eq. 21 against the five baseline models

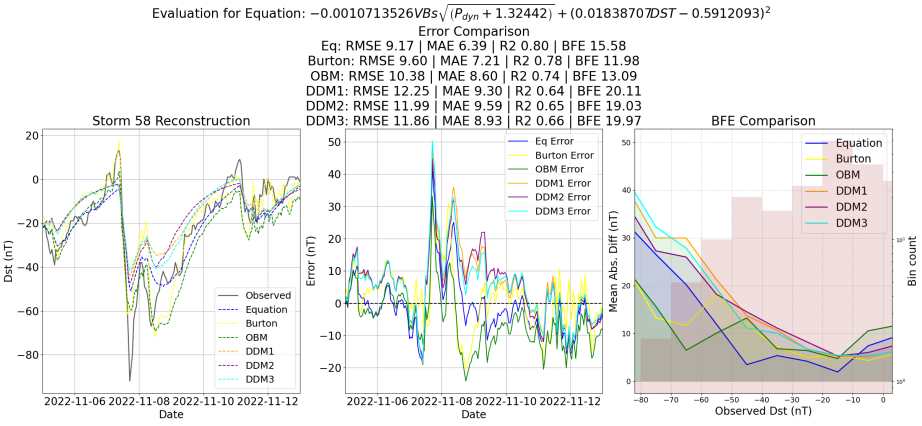

Figure 85: Reconstruction of storm 58 using Eq. 21 against the five baseline models

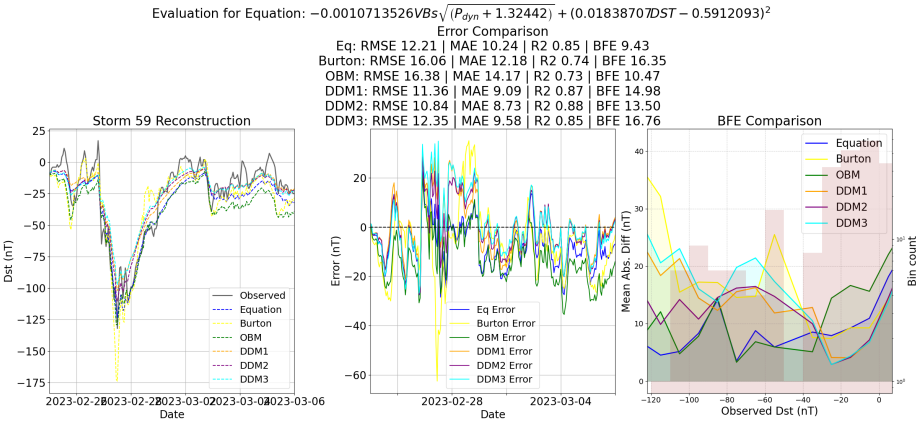

Figure 86: Reconstruction of storm 59 using Eq. 21 against the five baseline models

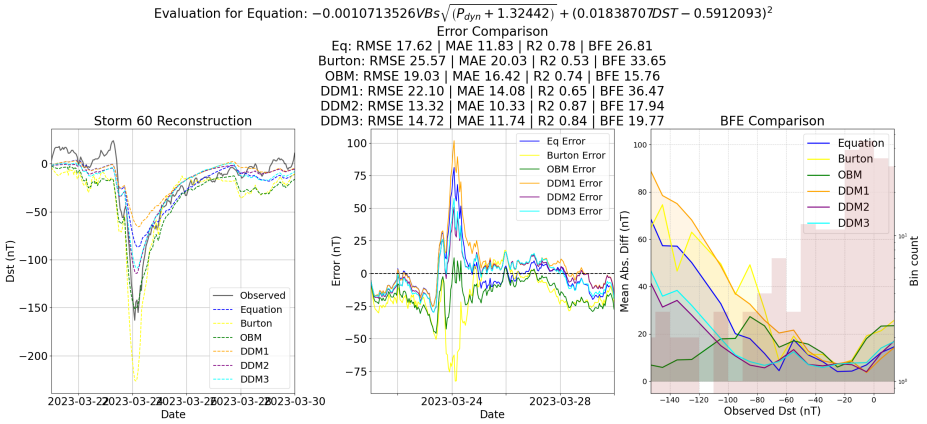

Figure 87: Reconstruction of storm 60 using Eq. 21 against the five baseline models

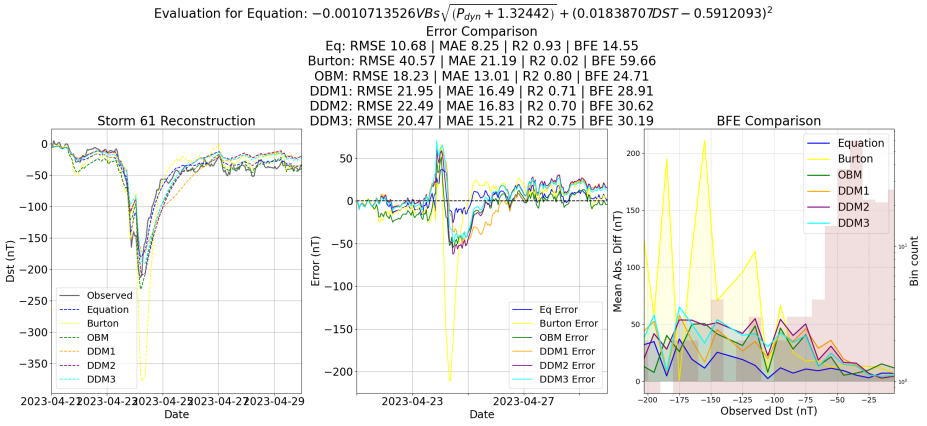

Figure 88: Reconstruction of storm 61 using Eq. 21 against the five baseline models

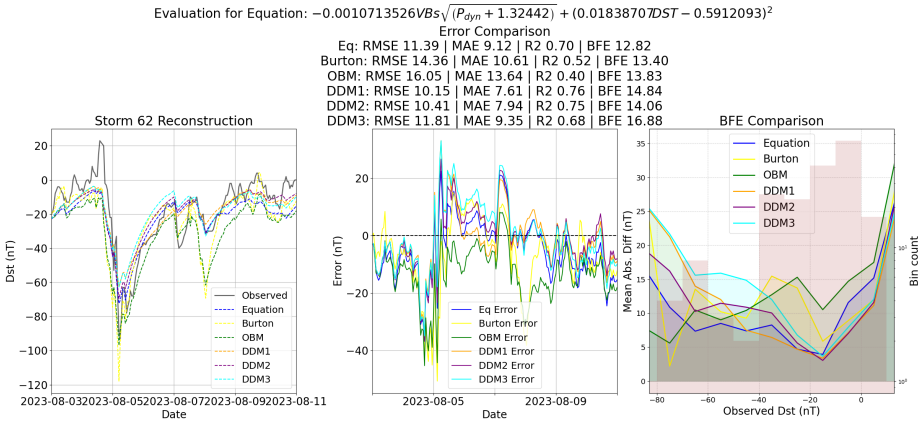

Figure 89: Reconstruction of storm 62 using Eq. 21 against the five baseline models

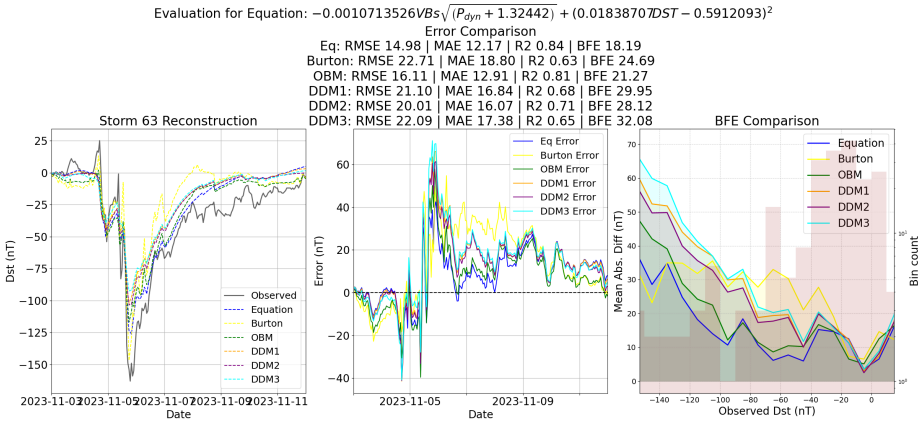

Figure 90: Reconstruction of storm 63 using Eq. 21 against the five baseline models

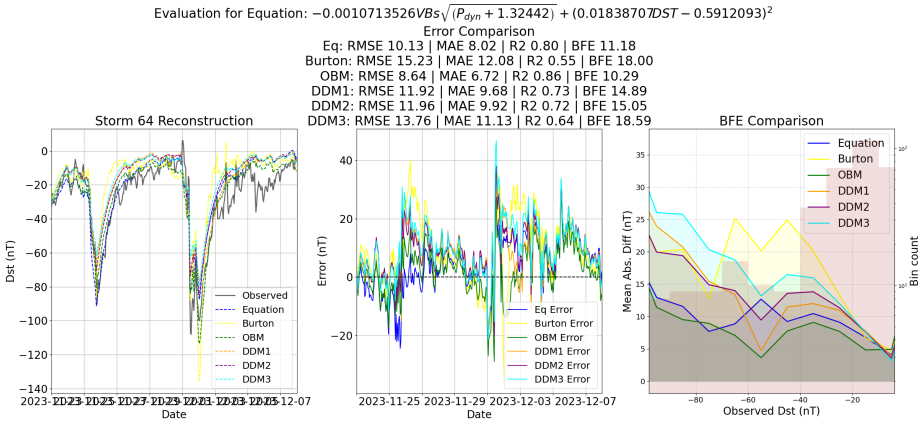

Figure 91: Reconstruction of storm 64 using Eq. 21 against the five baseline models

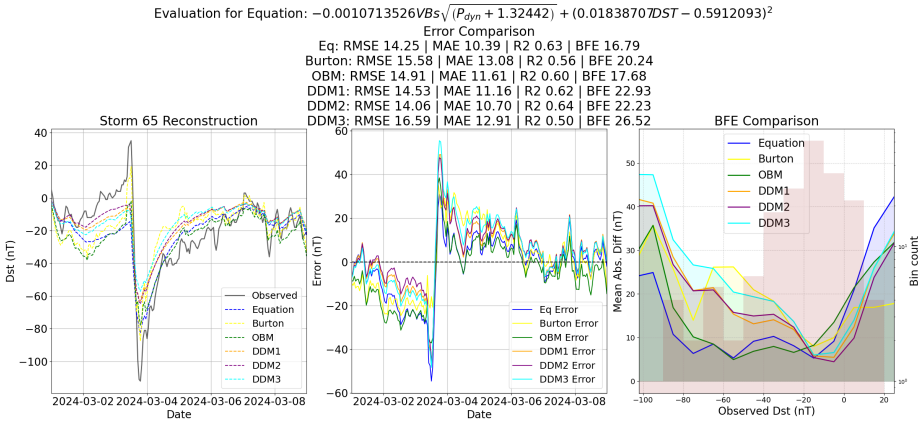

Figure 92: Reconstruction of storm 65 using Eq. 21 against the five baseline models

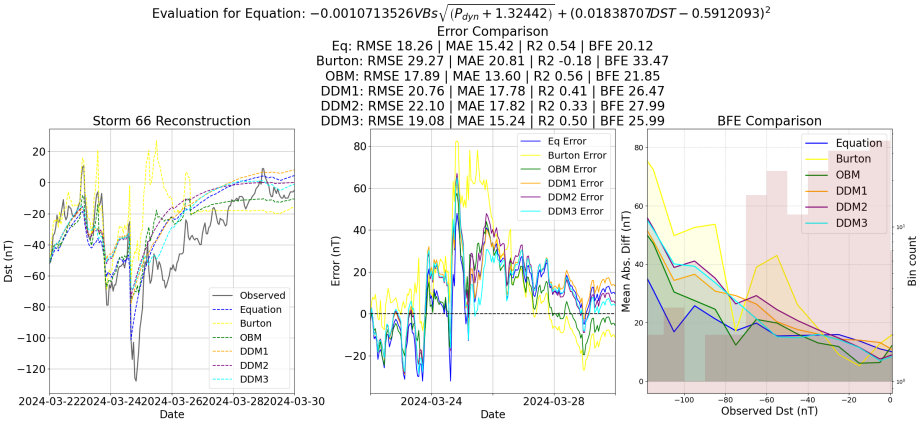

Figure 93: Reconstruction of storm 66 using Eq. 21 against the five baseline models

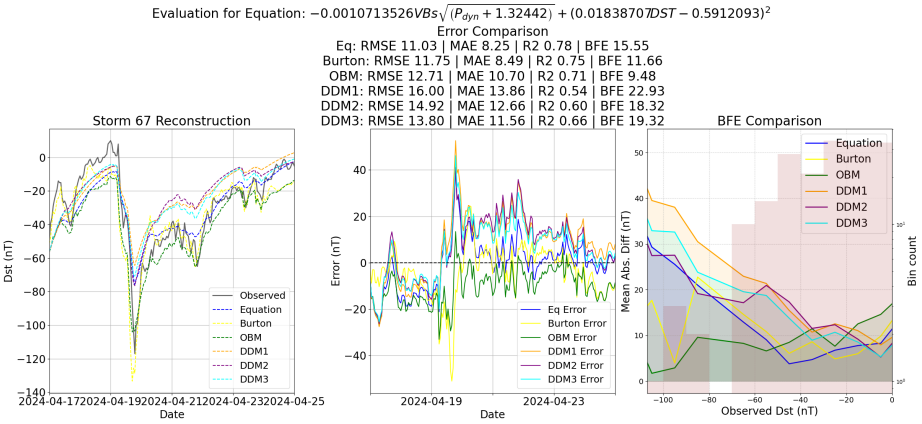

Figure 94: Reconstruction of storm 67 using Eq. 21 against the five baseline models

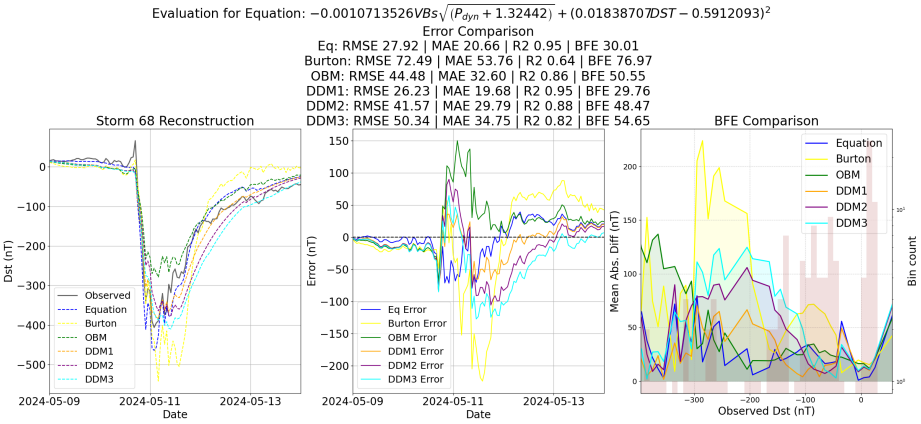

Figure 95: Reconstruction of storm 68 using Eq. 21 against the five baseline models

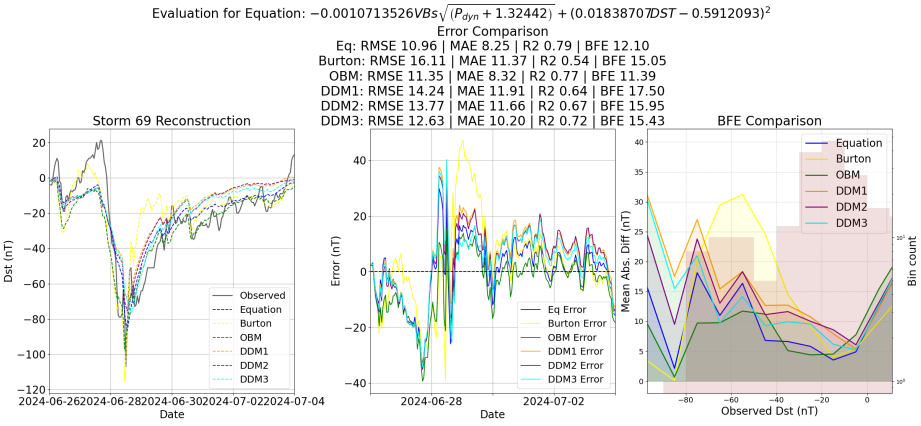

Figure 96: Reconstruction of storm 69 using Eq. 21 against the five baseline models

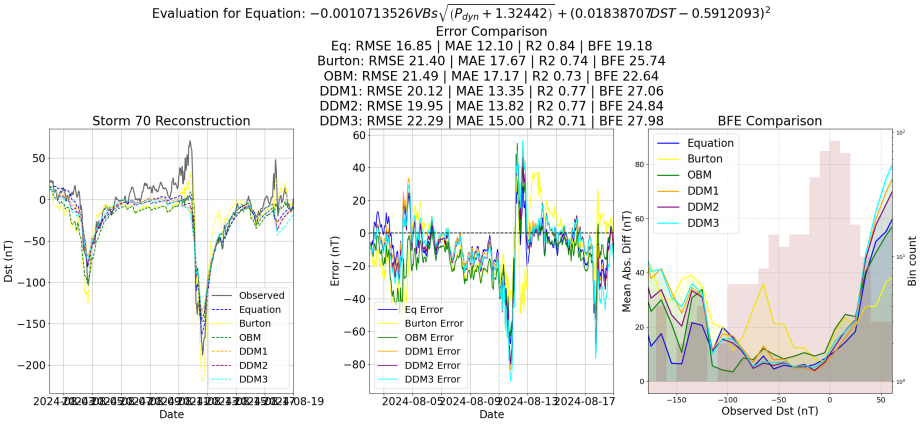

Figure 97: Reconstruction of storm 70 using Eq. 21 against the five baseline models

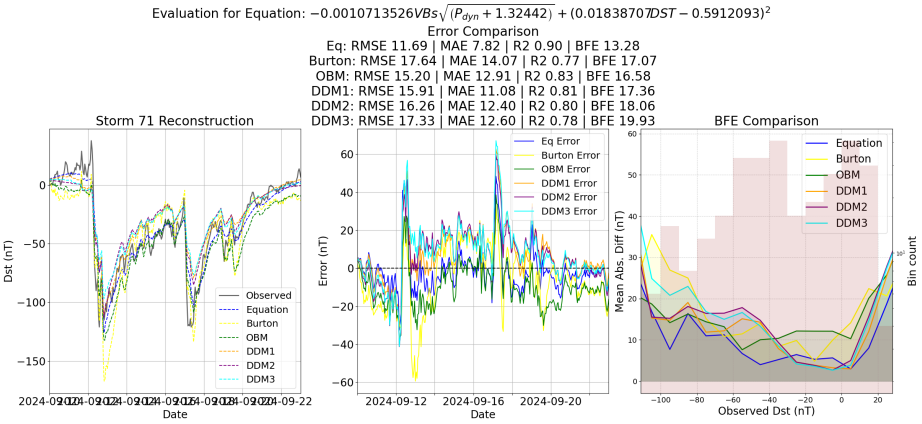

Figure 98: Reconstruction of storm 71 using Eq. 21 against the five baseline models

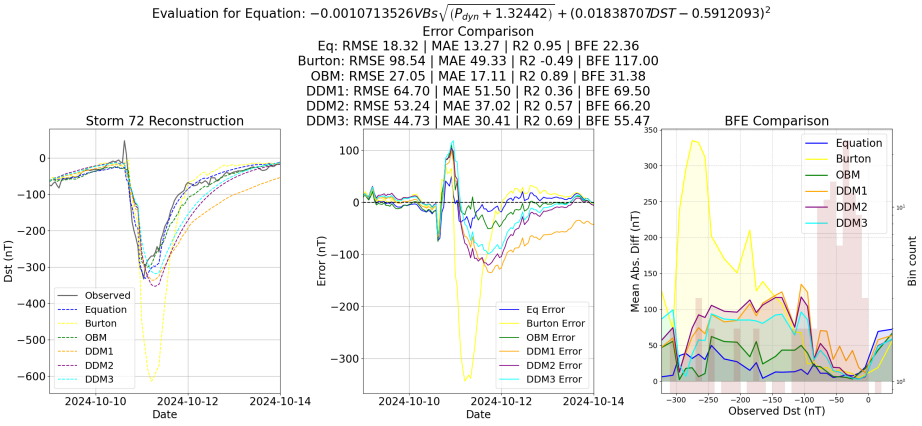

Figure 99: Reconstruction of storm 72 using Eq. 21 against the five baseline models

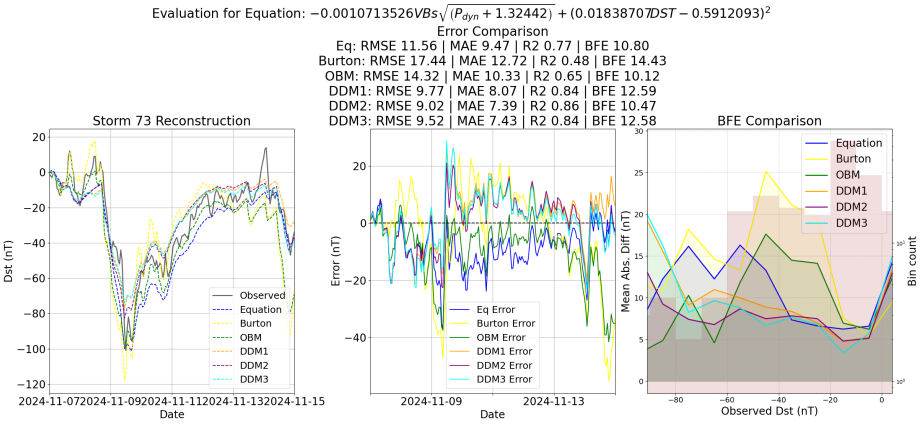

Figure 100: Reconstruction of storm 73 using Eq. 21 against the five baseline models